Lezione 1
introduction & schematic

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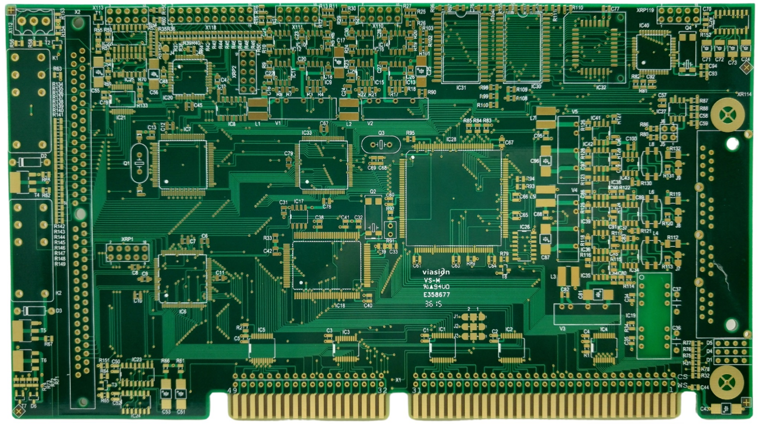
13/01/2026

Course Objectives

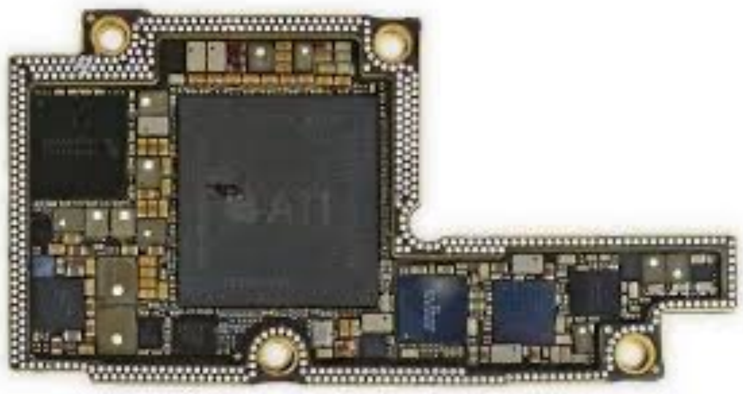
- Manage hardware projects using **Git version control**
- Design **PCB schematics and layouts** using **KiCad**
- Perform **basic SPICE simulations** to validate critical circuits
- Understand **FPGA architecture and HDL design methodology**
- Develop, simulate, and synthesize simple **VHDL modules**
- Use **XILINX VIVADO** and integrate IP cores

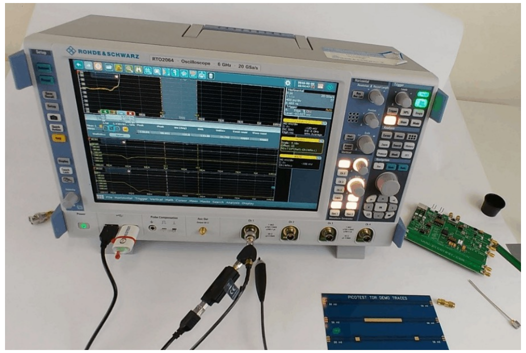
PCB, electronics

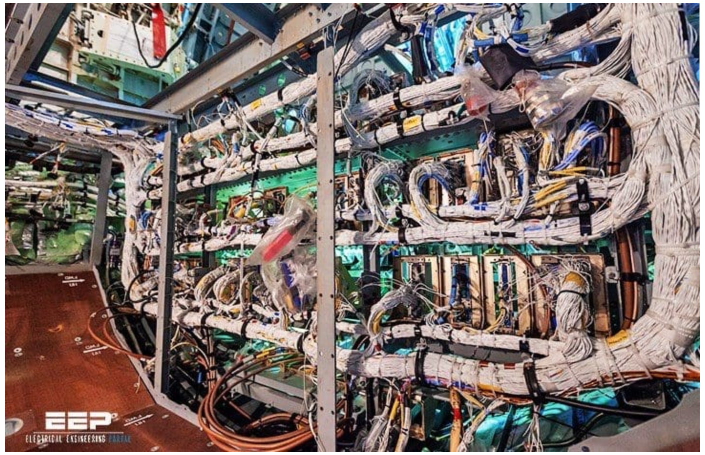


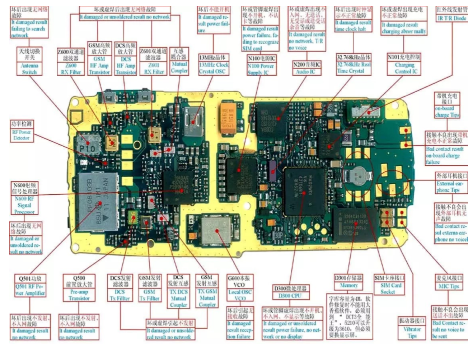
- Consumer Electronics
- Industrial & Automotive
- Telecommunications & Networking
- Medical Electronics
- Aerospace, Defense, & Maritime
- Lighting
- Instrumentation

Ecc ….





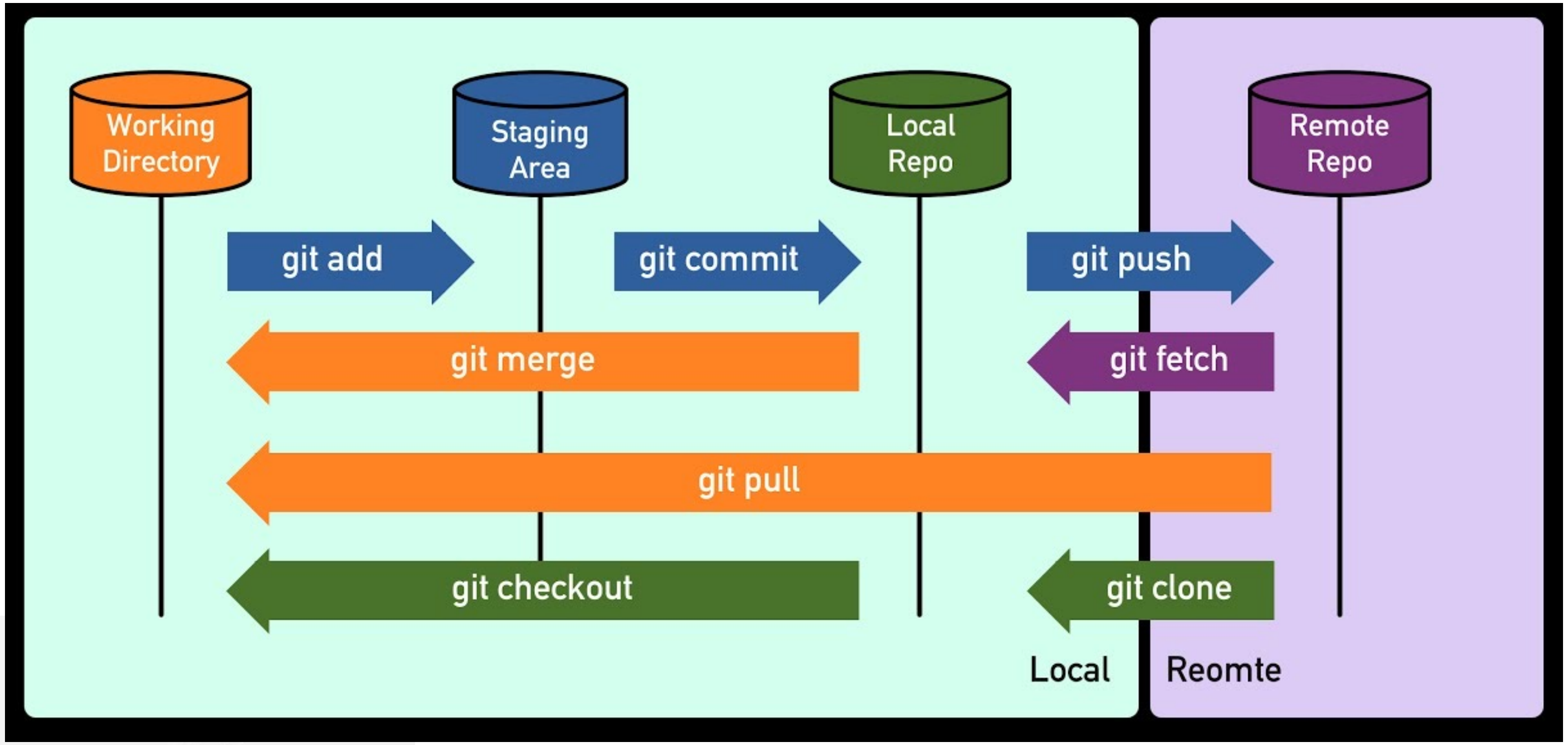




Tools Used

- Git (versioning)
- Draw.io (block diagrams)
- KiCad (Schematic & PCB)
- NGspice & LTspice (simulation)
- XILINX VIVADO (FPGA)

Git


Working Directory
Staging Area
Local Repo
Remote Repo
git add
git commit
git push
git merge
git fetch
git pull
git checkout
git clone
Local
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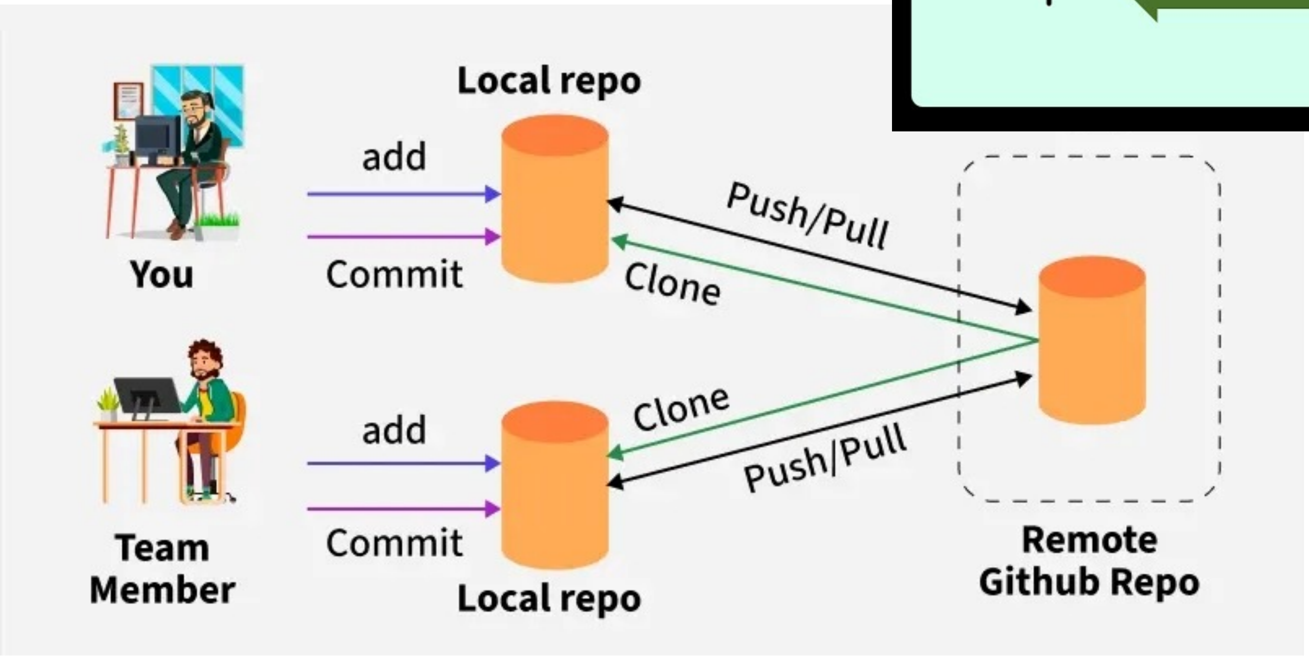
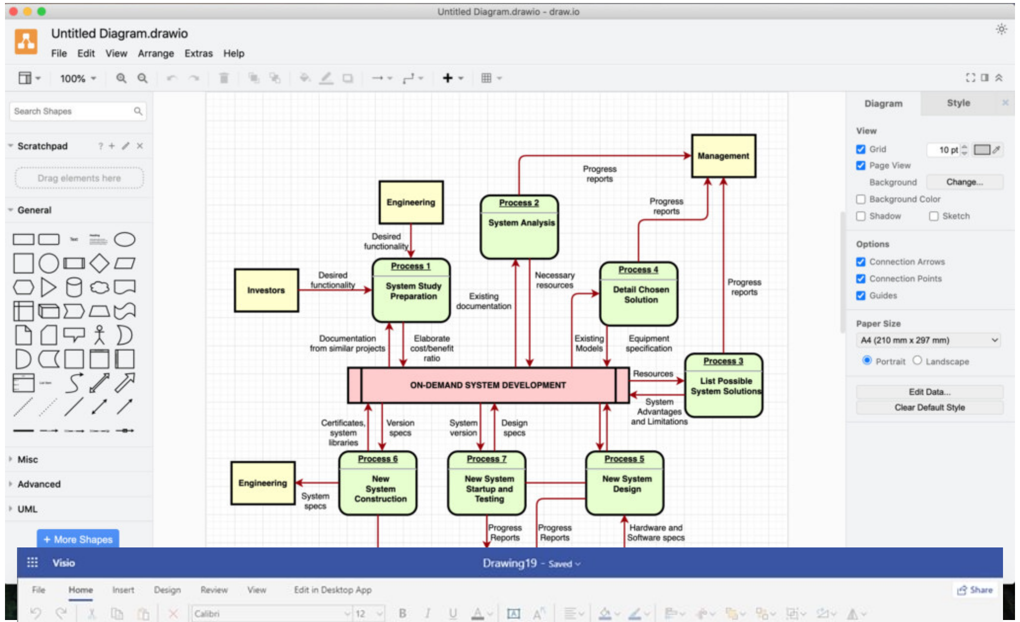
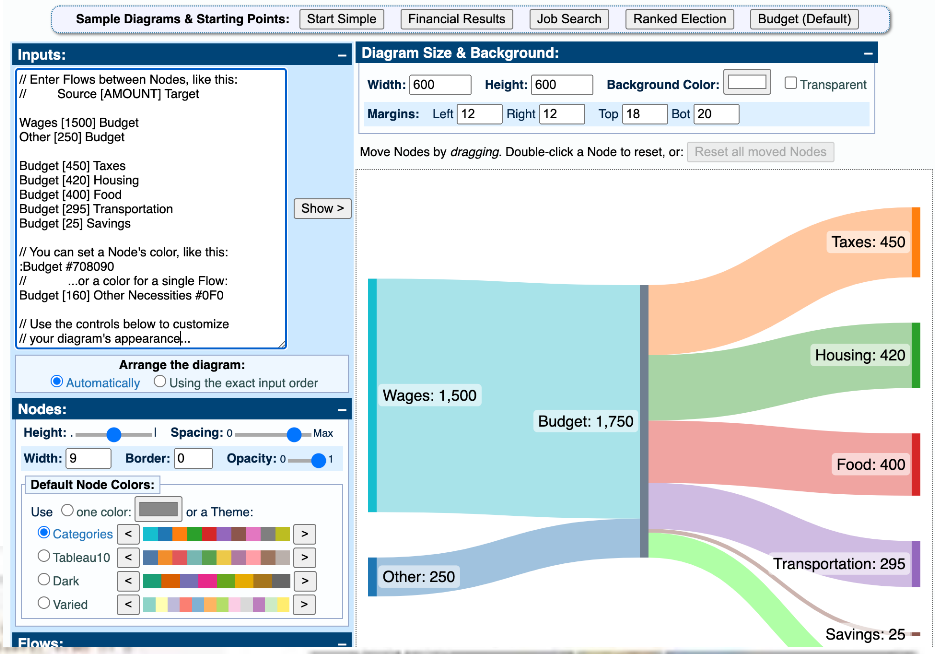
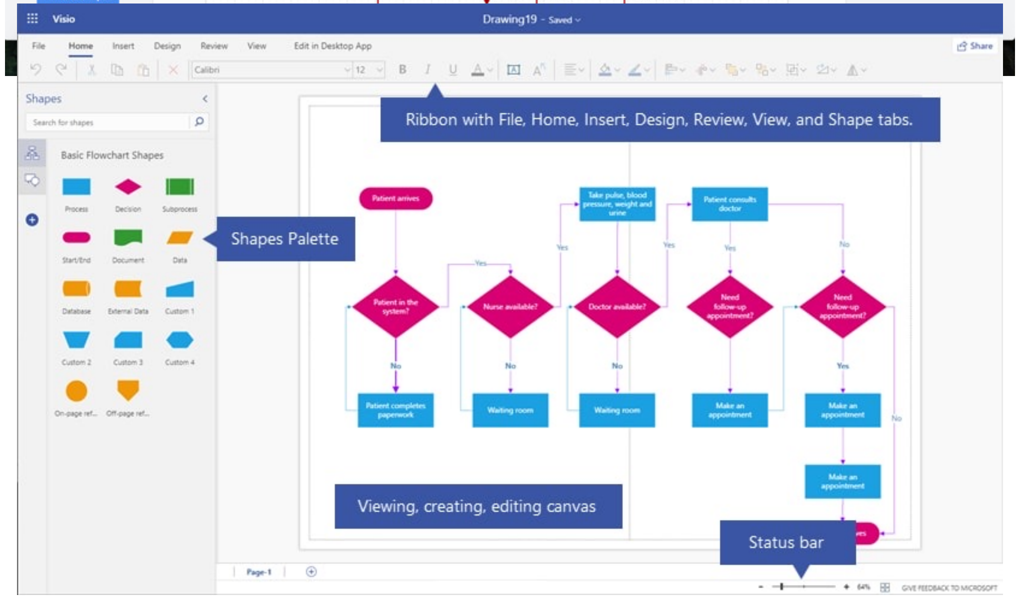
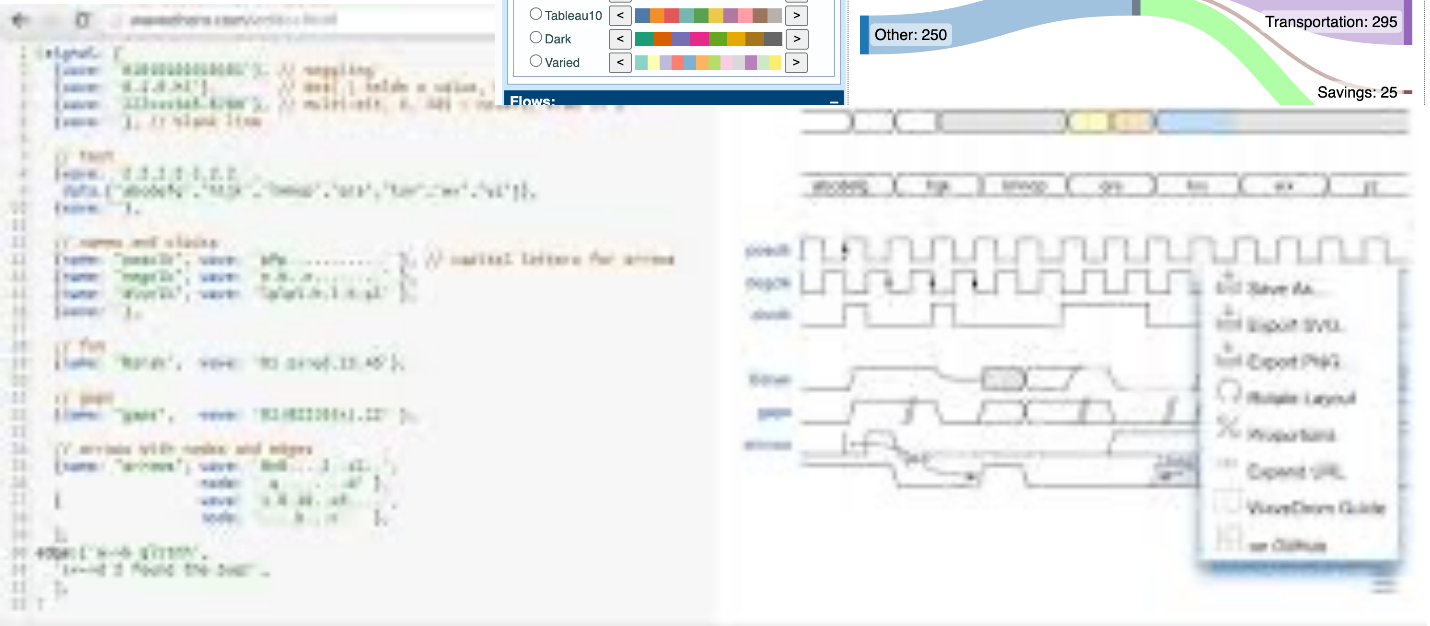

You
Local repo
add
Commit
Push/Pull
Clone
Team Member
add
Commit
Local repo
Clone
Push/Pull
Remote Github Repo

Draw.io – wavedrom – sankeymatic – MS Visio









Major Electronic Design Automation (EDA) vendors

xpedition

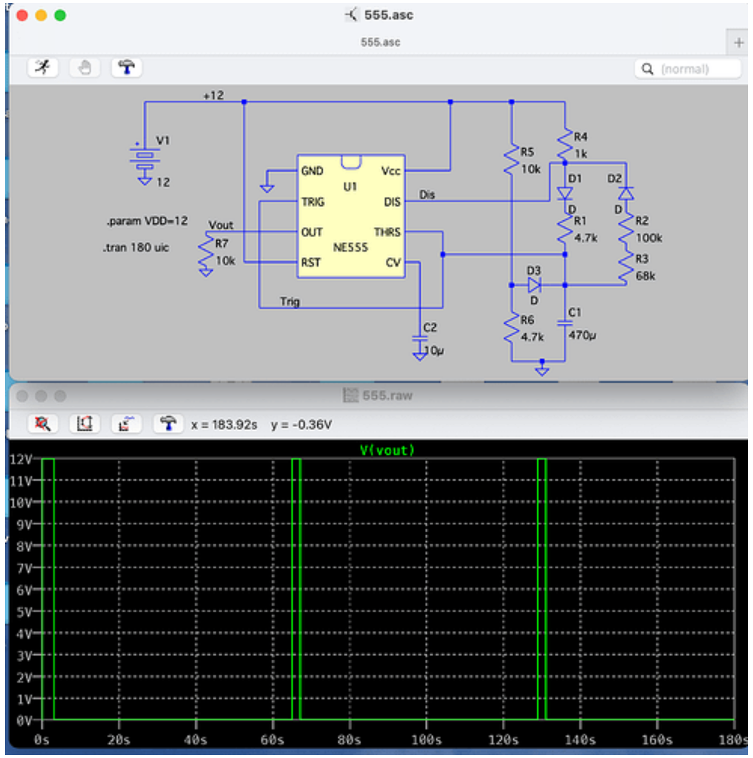
Altium Designer

Mentor®
A Siemens Business

KiCad

cādence®
Allegro®

SPICE simulation

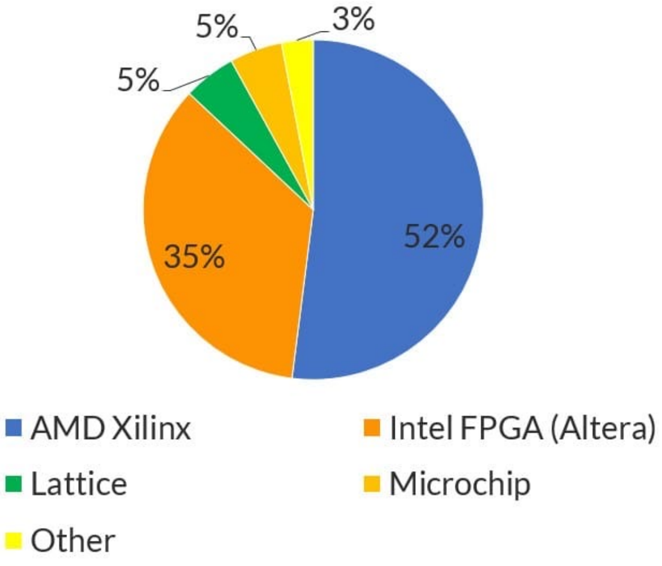


SIMULATION

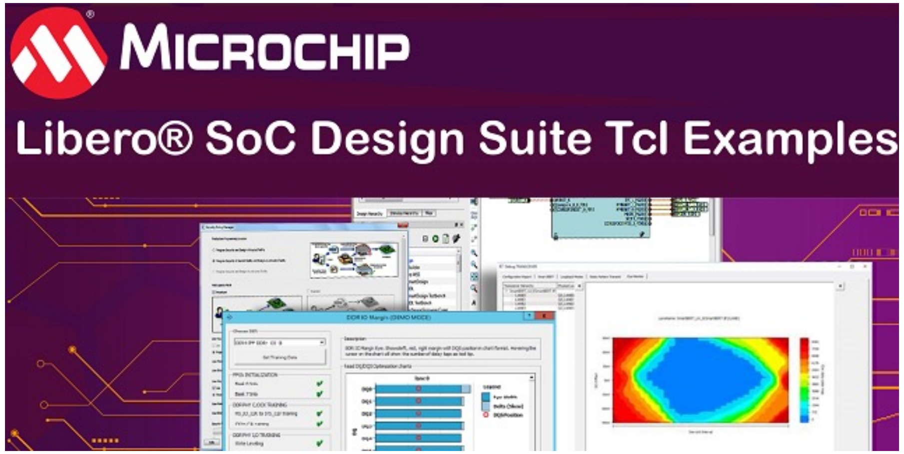
LTSPice v/s PSpice

Parameter	Linear Technology's SPICE	Personal SPICE
Cost & Licensing	Completely free (even for commercial use)	Paid (Full version); free version is limited
Simulation Speed (Switching Circuits)	Extremely fast, optimized for SMPS & analog switching	Slower in power circuits, can require convergence tweaks
User Interface & Learning Curve	Lightweight UI, scripting-based setup	Advanced GUI with easier access to features
Analog vs Digital Support	Strong analog focus, minimal digital logic	True mixed-signal with full digital gate library
Monte Carlo & Tolerance Analysis	Supported via scripting (`.step`, `gauss()`) – powerful but manual	Native built-in support with statistical tools
Integration with PCB Design Tools	No native PCB integration	Fully integrated with OrCAD/Allegro (Cadence Suite)
Model Library Support	Best with Analog Devices models; 3rd party models need tweaks	Wide vendor-neutral support (TI, ST, NXP, ON Semi, etc.)
Simulation Features (Advanced)	Supports behavioral modeling, parametric sweeps, some noise	Adds sensitivity, reliability, yield analysis, fault simulation
Best Use Case Fit	Analog design, SMPS, filters, discrete validation	Full system-level sim (MCU + analog + bus-level behavior + PCB)

FPGA design suite



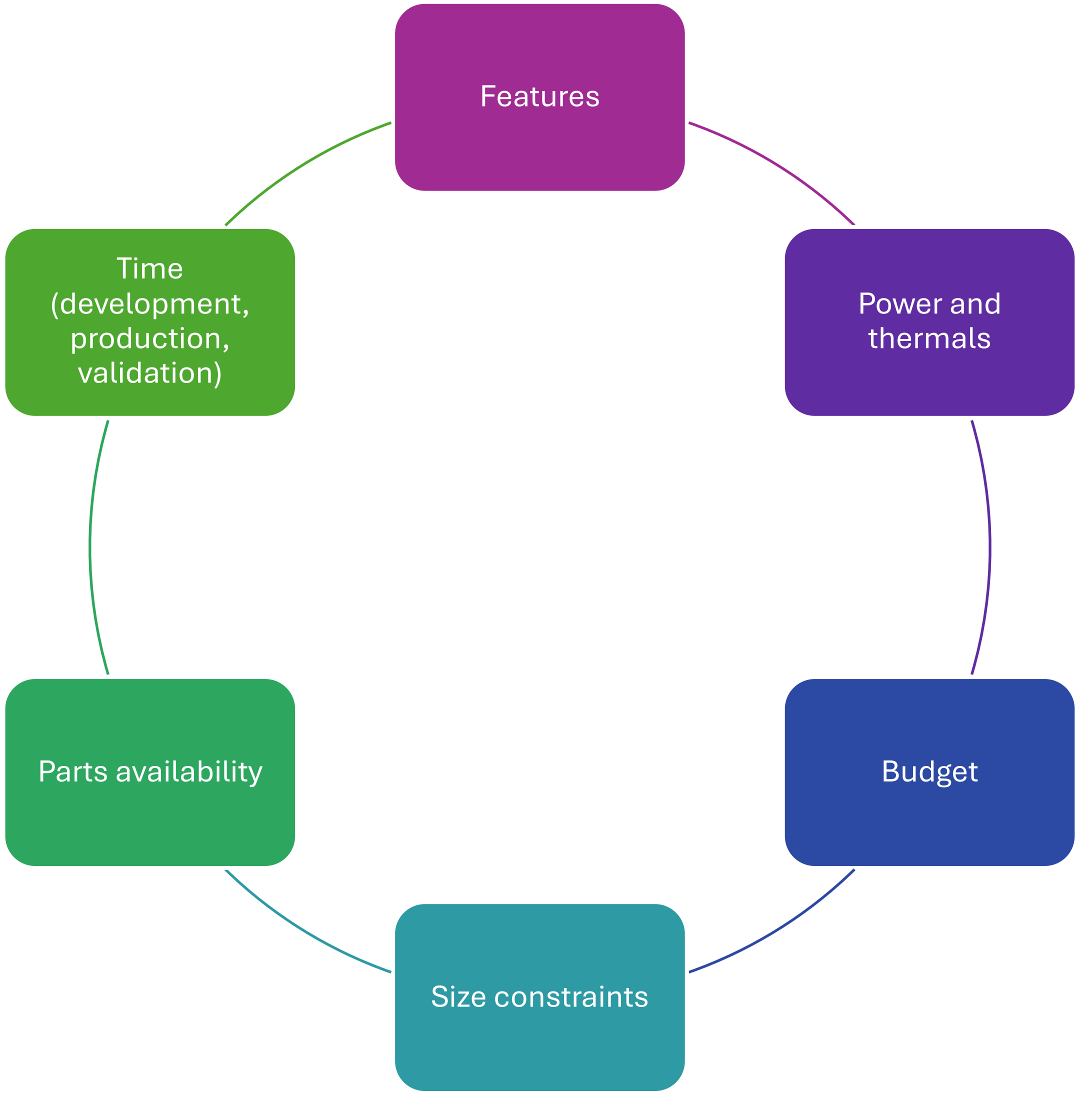
Source: The Information Network

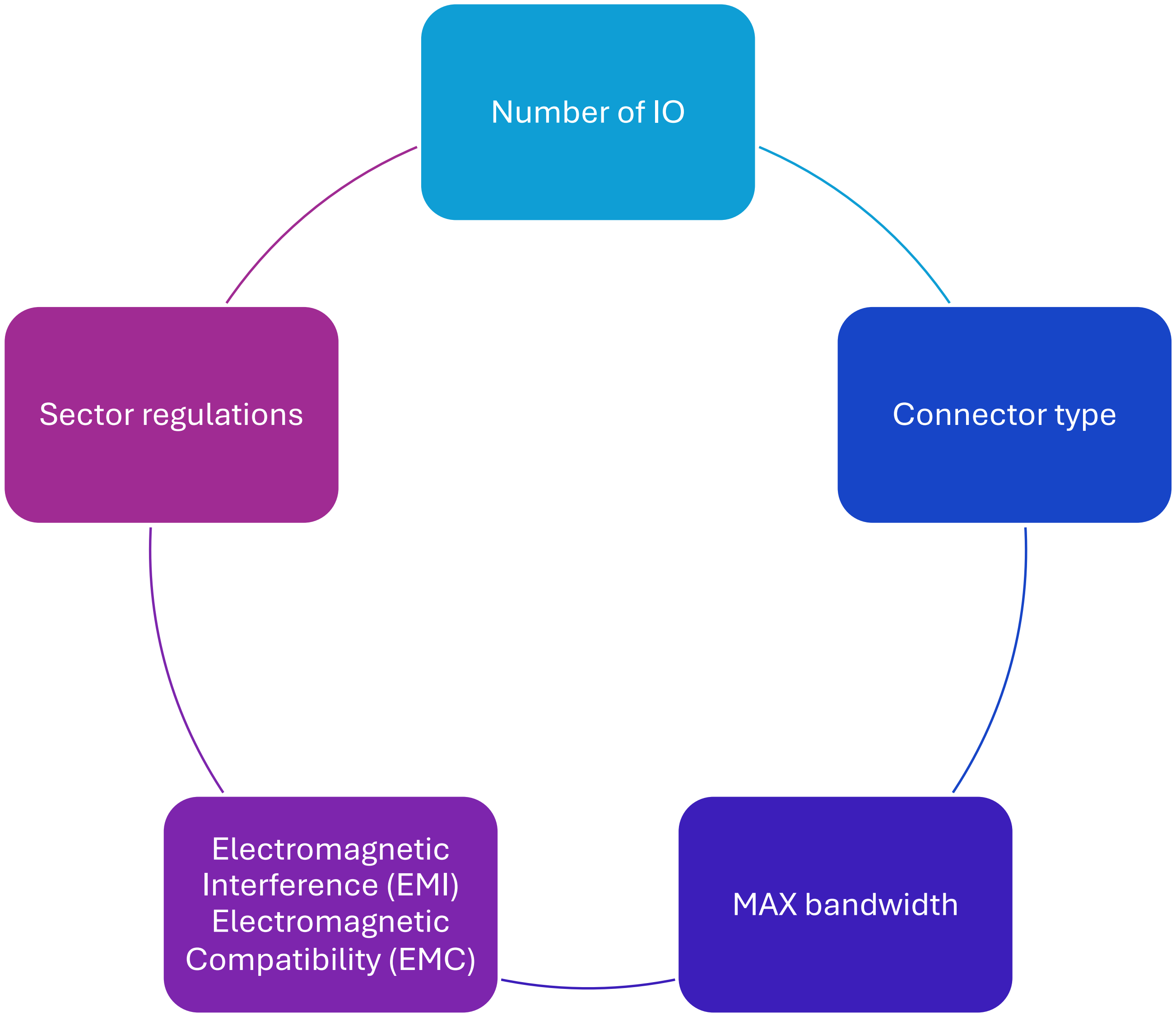


PCB design process

1. Gather requirements
2. Block diagram
3. Schematic Design
4. Simulation
5. Bill Of Materials
6. PCB Layout
7. (Simulation)
8. Manufacturing
9. Testing and validation

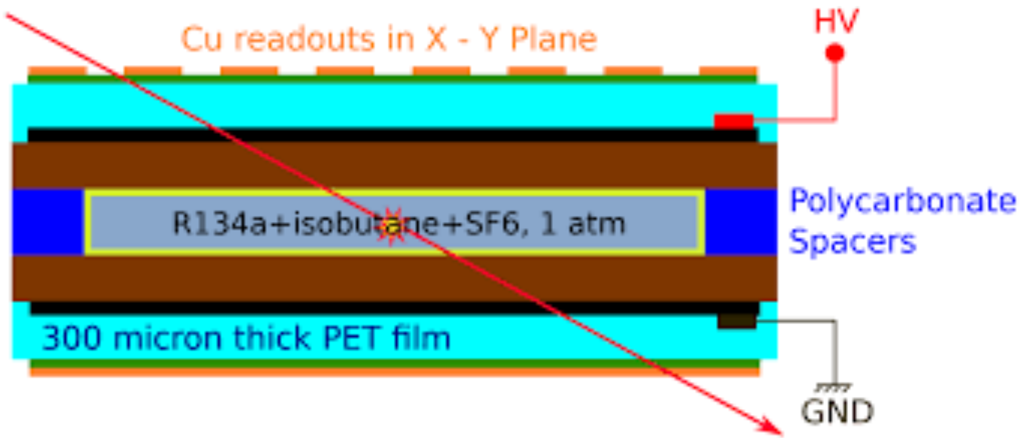
Gather requirements





Course goal (Gather requirements)

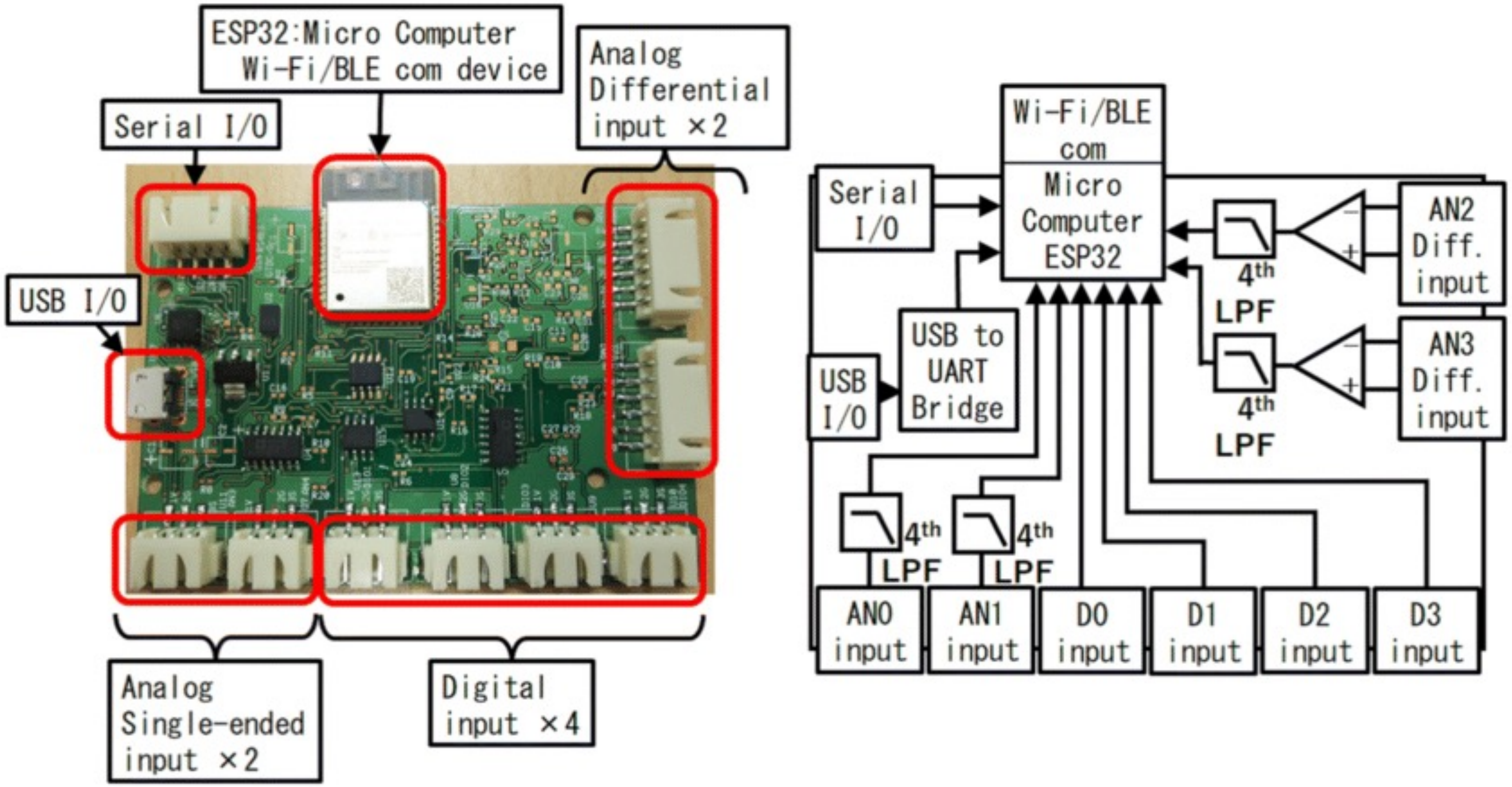
Resistive Plate Chamber (RPC) diagram



Develop a custom module used to detect currents from 5 uA to 700 uA.

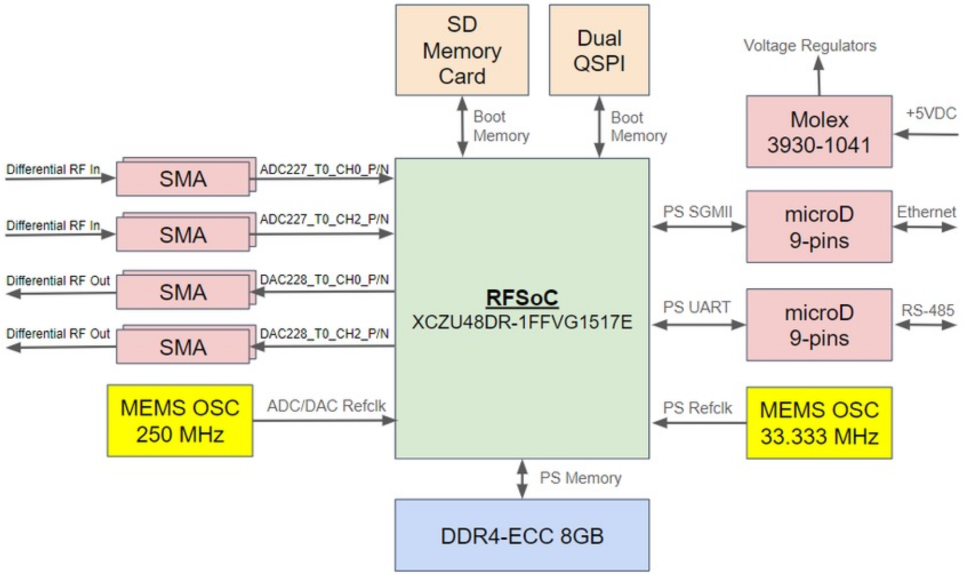
- **8 inputs connected to a 16 bit ADC (8 to 860 SPS)**
- **Shunt resistor with Current-Sense Amplifier**
- **20 KHz -3 dB FEE bandwidth low pass filtering**
- **Microcontroller data taking with of the shelf module.**
- **Low noise**
- **FEE insulated from digital section**

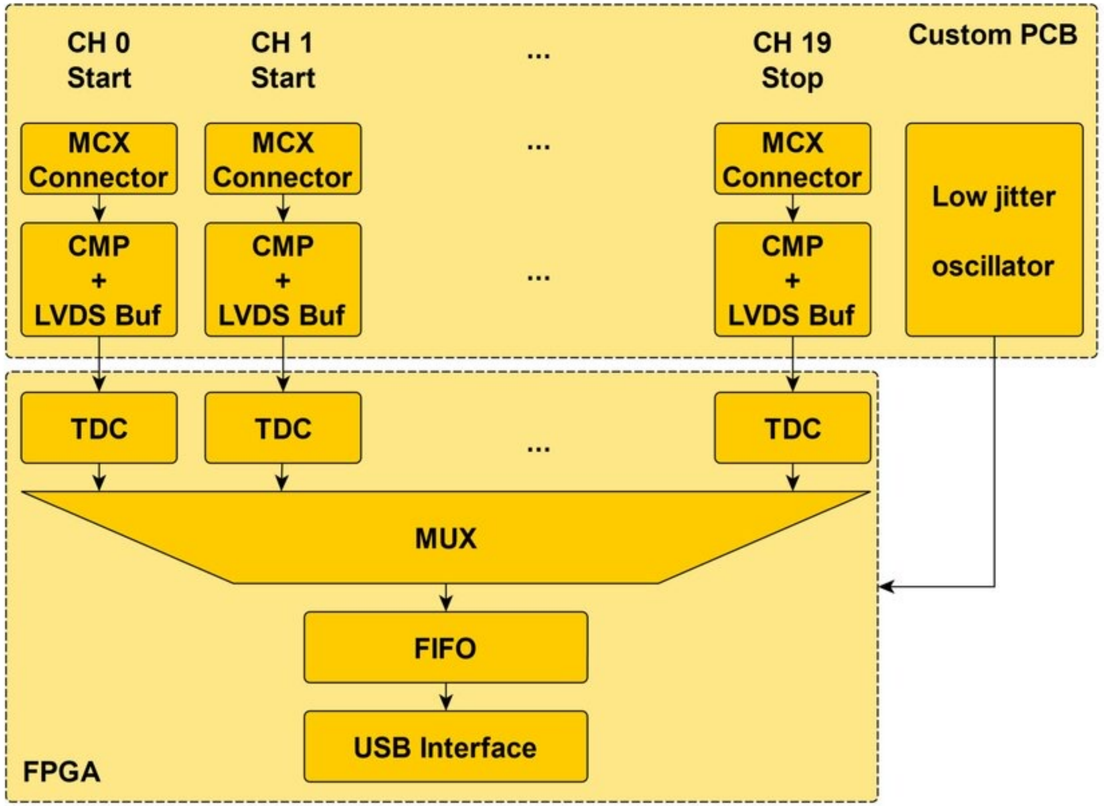
Block diagram – 1



(a)PCB appearance (b)Block diagram

Block diagram – 2

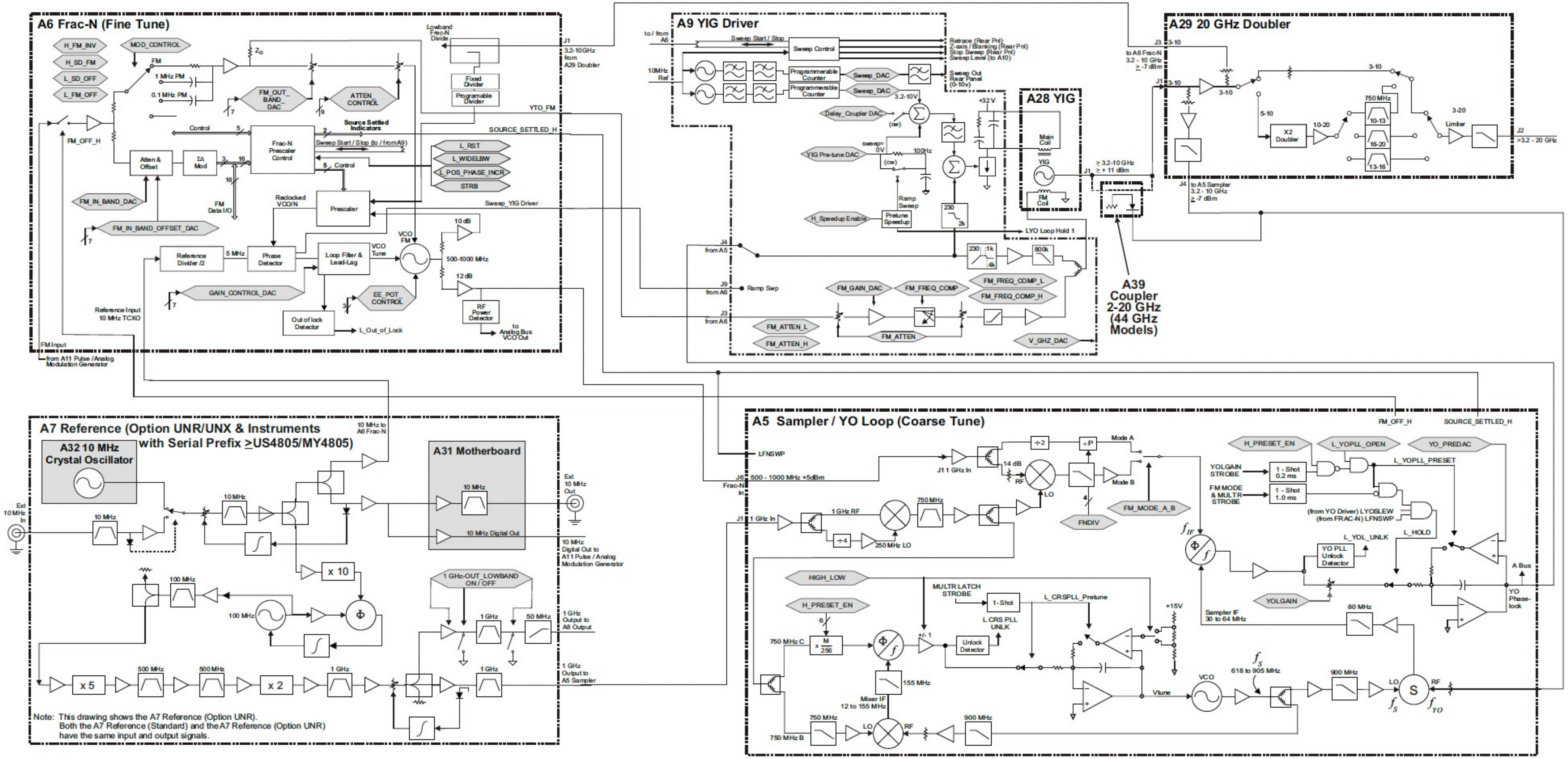




Agilent E8257D block diagram example

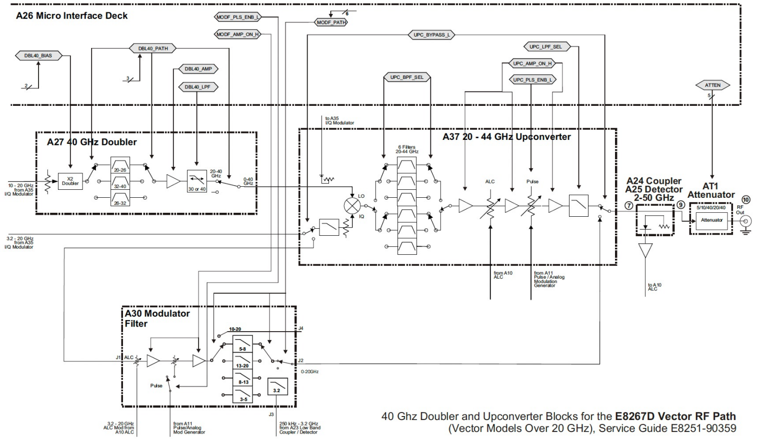
Agilent Technologies
E8257D/67D, E8663D
PSG Signal Generators

Service Guide

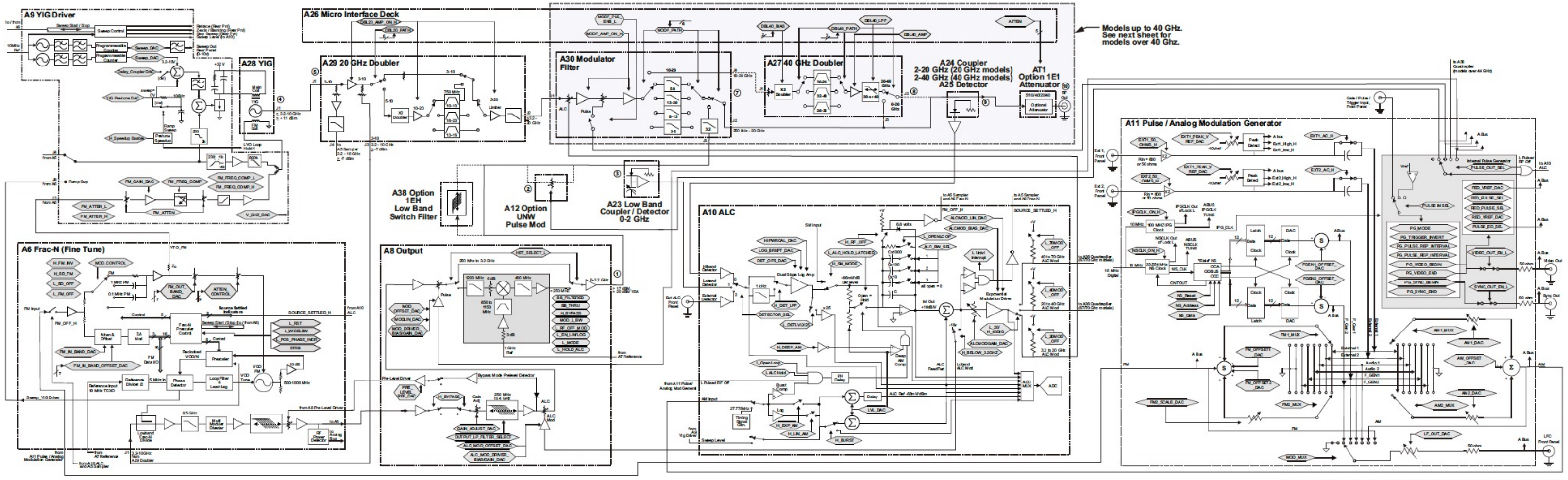


Block Diagram for the E8257N Generator, Sweep Reference / Synthesis Loop, Service Guide E8251-90014

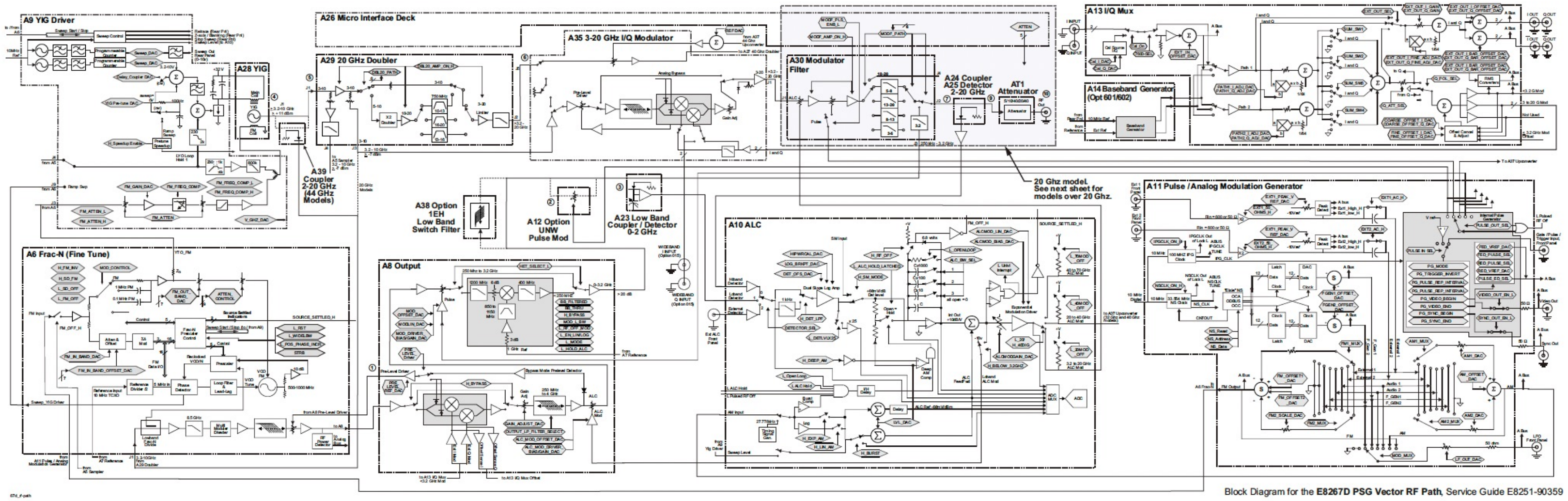
Agilent E8257D block diagram example



40 Ghz Doubler and Upconverter Blocks for the **E8267D Vector RF Path** (Vector Models Over 20 GHz), Service Guide E8251-90359

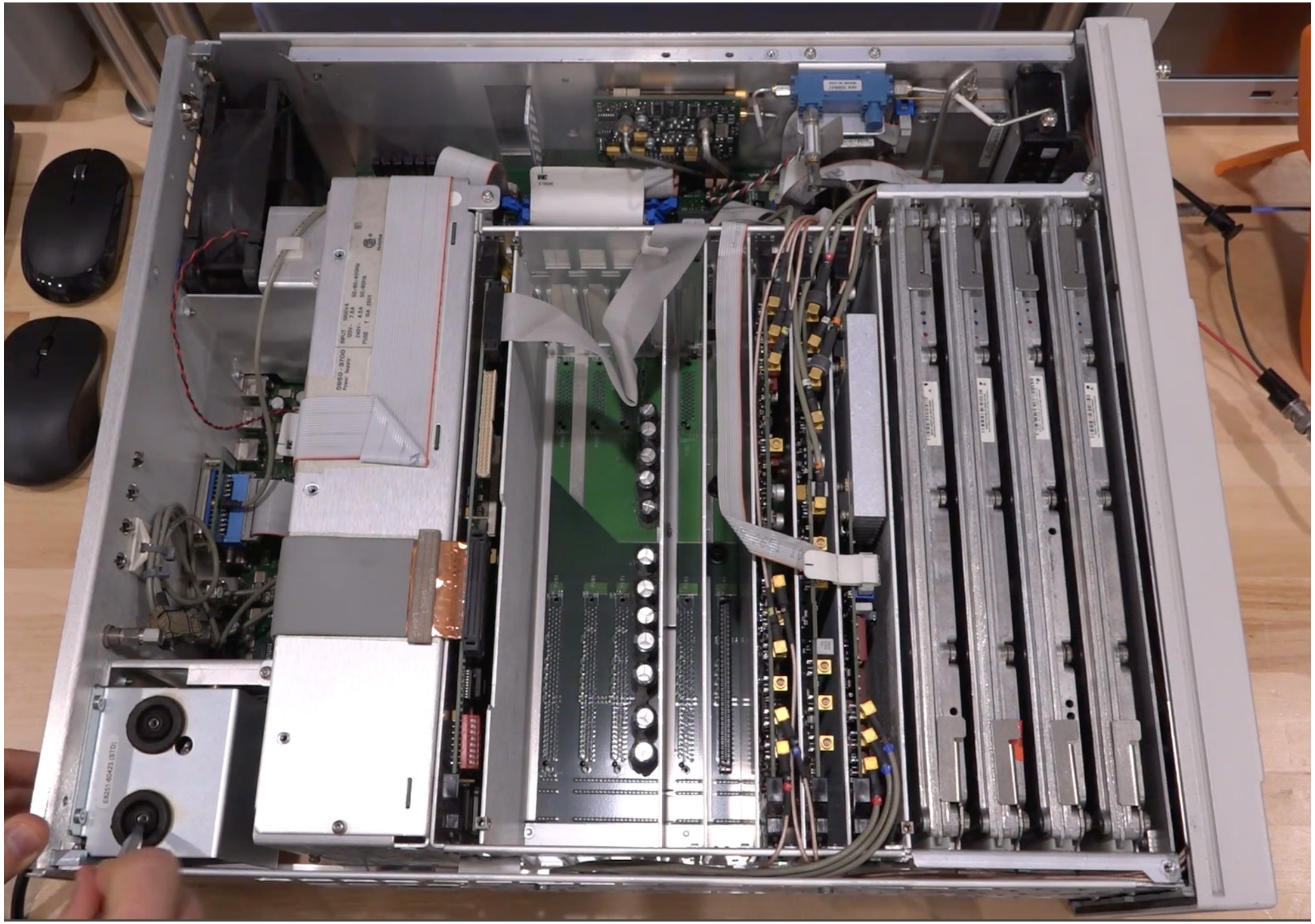


Block Diagram for the **Analog RF Path**, Service Guide E8251-90359



Block Diagram for the **E8267D PSG Vector RF Path**, Service Guide E8251-90359

Agilent E8257D



Hands-on session 0 – Git

“A distributed system to track changes in source code, enabling multiple people to work together on non-linear development.”

Setup & Basics

`git init` - Start repo

`git clone <url>` - Copy repo

`git add .` - Stage all

`git commit -m "msg"` - Save

`git push` - Upload

`git pull` - Download

Branch & Merge

`git branch` - List branches

`git checkout <branch>` - Switch

`git merge <branch>` - Combine

`git branch -d <name>` - Delete

Investigate & Fix

`git log` - View history

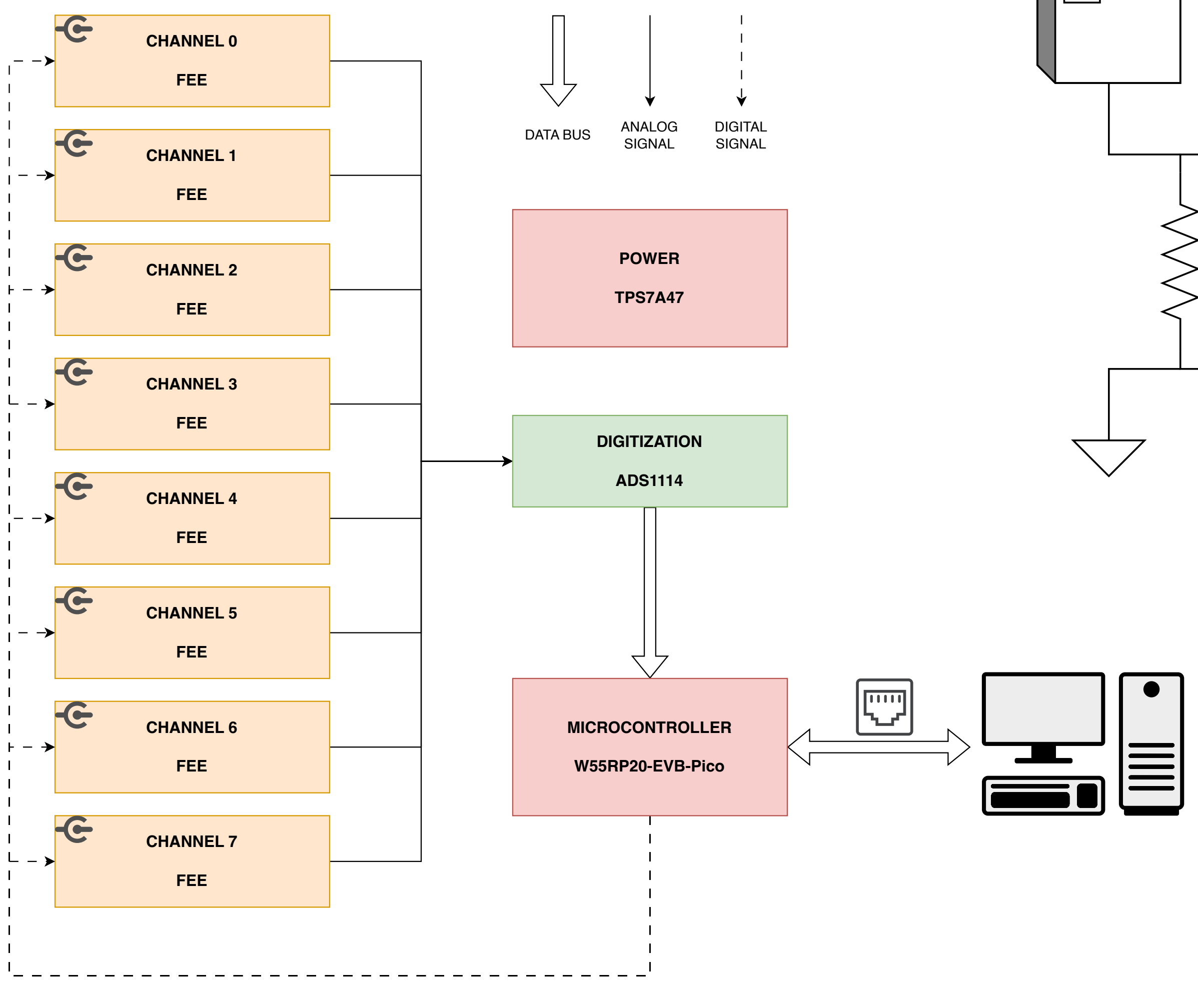
`git status` - Check changes

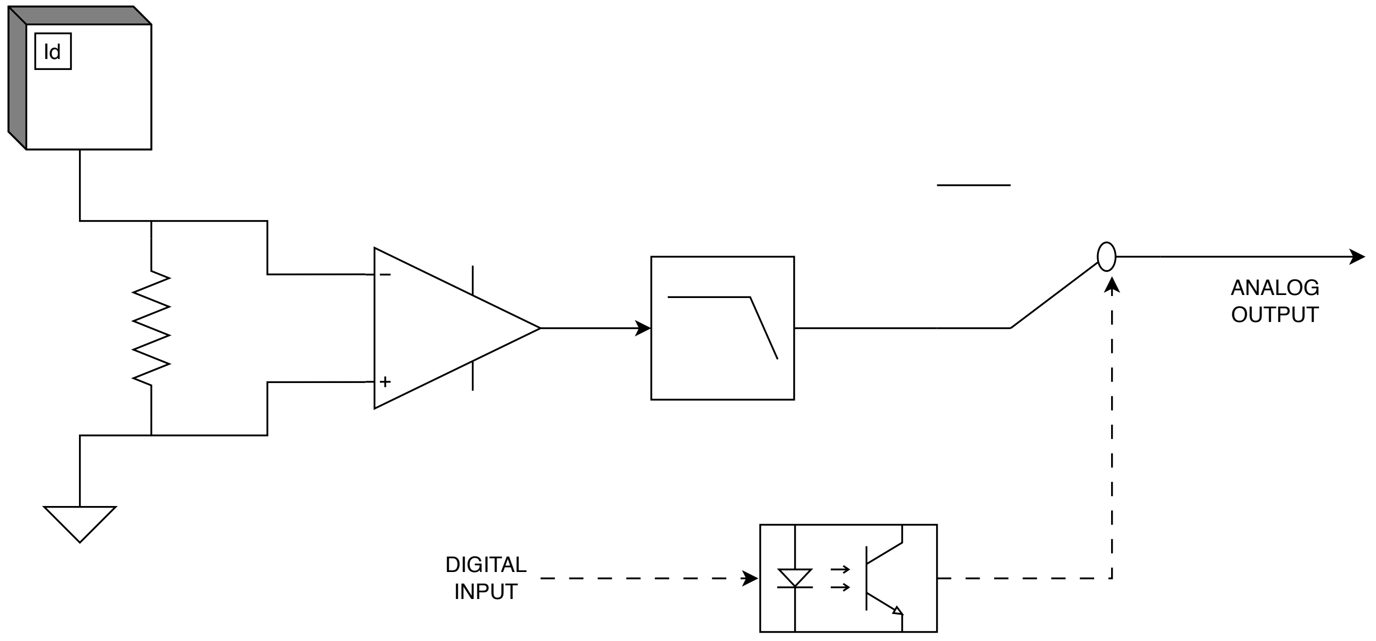
`git diff` - See differences

`git restore <file>` - Undo changes

`git stash` - Temporarily save

Hands-on session 1 – Draw.IO

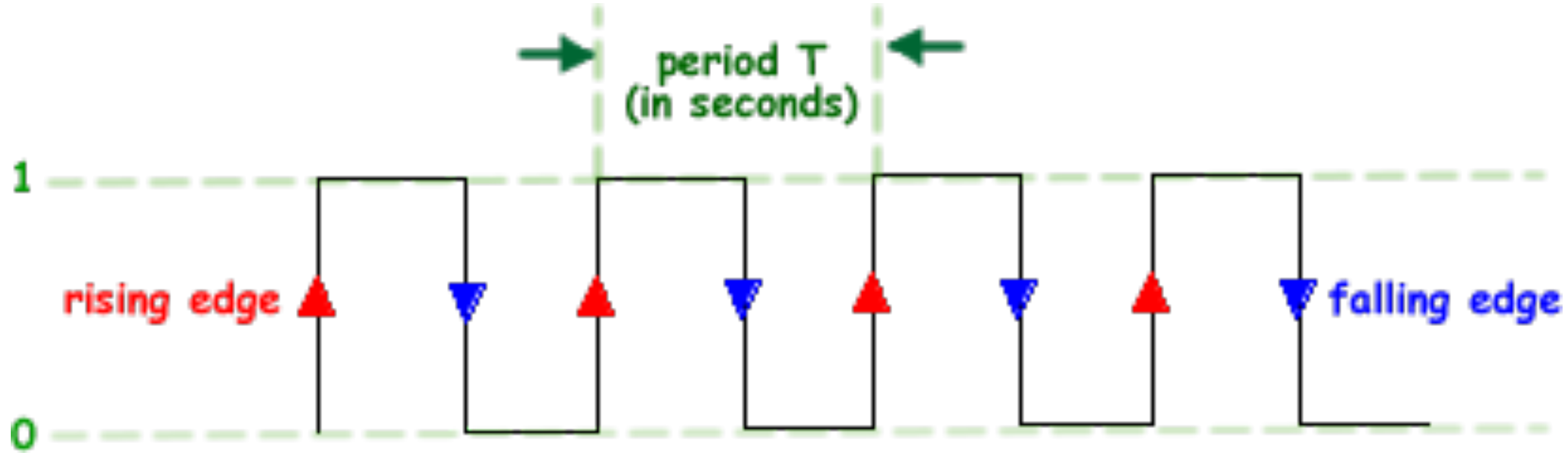




A bit of nomenclature

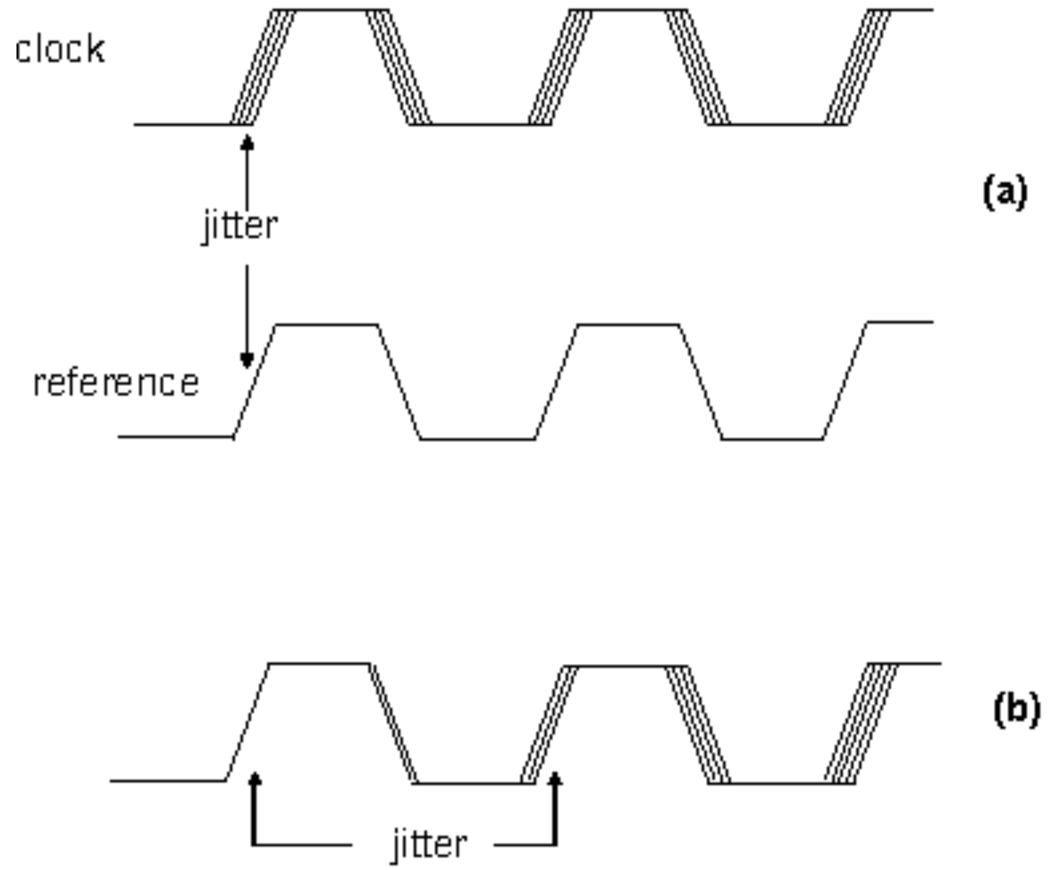
Clock

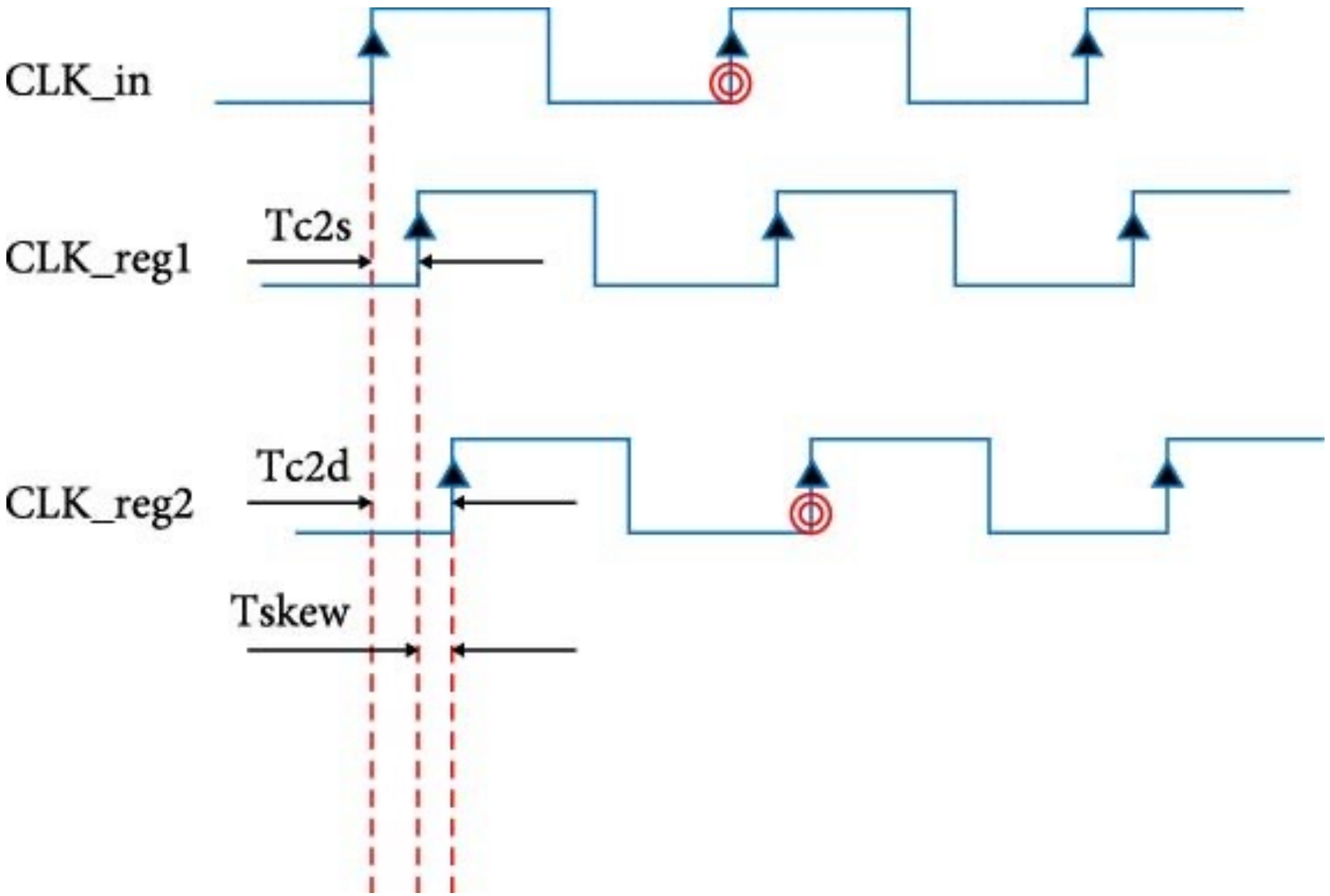
A **clock** is a periodic timing signal used to synchronize operations in an electronic system, defining when data is sampled, transferred, or processed.



- **62.5 ns (16 MHz):** Arduino UNO
- **20.8 ns (48 MHz):** USB full-speed devices and many microcontrollers
- **10 ns (100 MHz):** Common system clock for digital logic and CPUs
- **1 ns (1 GHz):** High-speed processors and modern communication systems

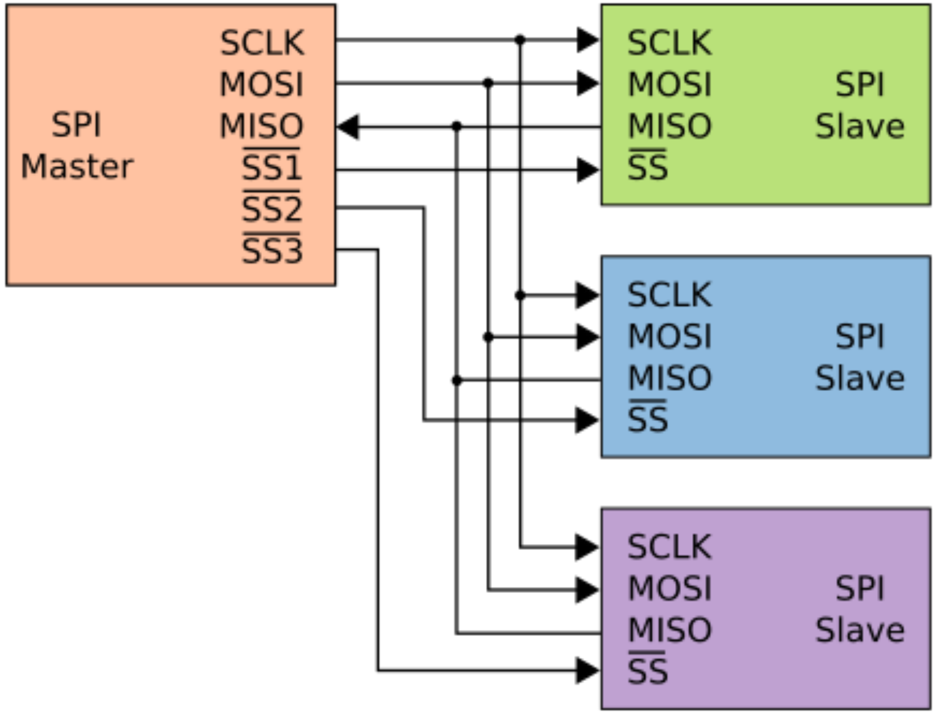
Skew and Jitter





- **Skew** is the fixed or slowly varying difference in timing between related signals (for example, two clock or data lines) that are intended to arrive simultaneously but do not due to path length, component, or loading differences.
- **Jitter** is the short-term, random or deterministic variation in the timing of a signal’s edges from their ideal positions, typically caused by noise, interference, or instability in the clock source.

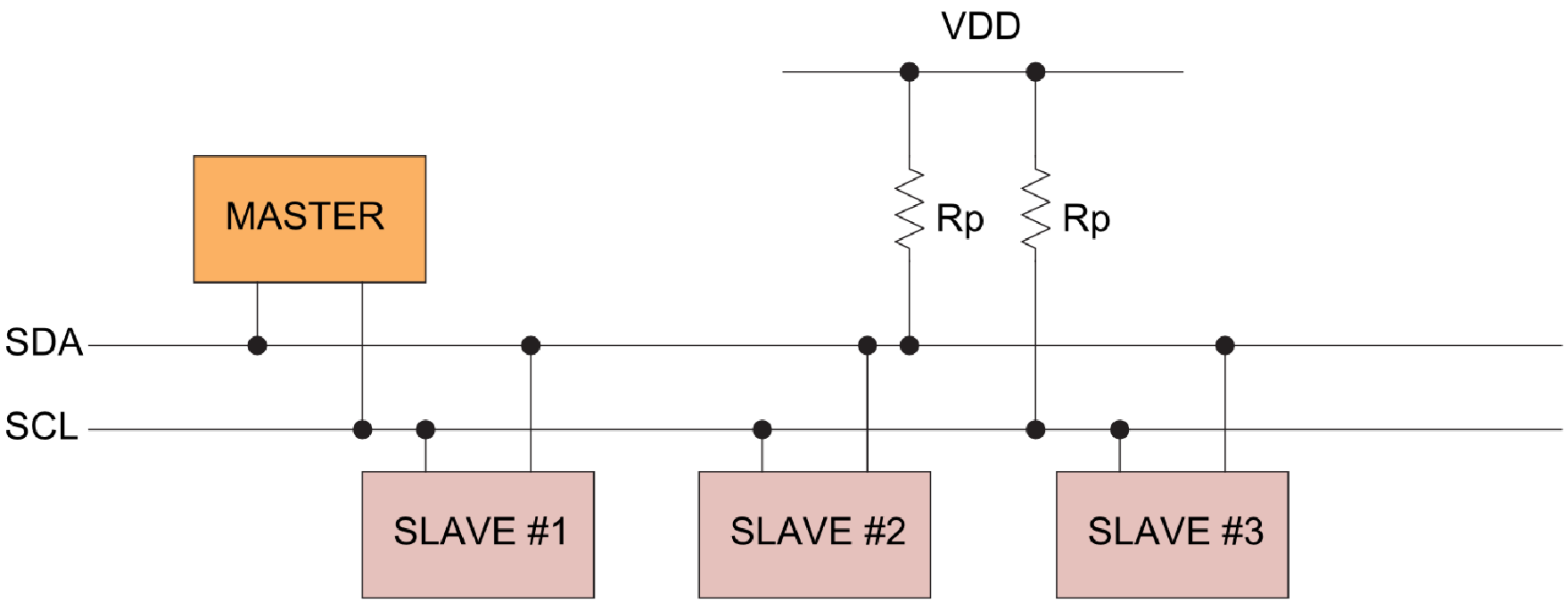
SPI (Serial Peripheral Interface)



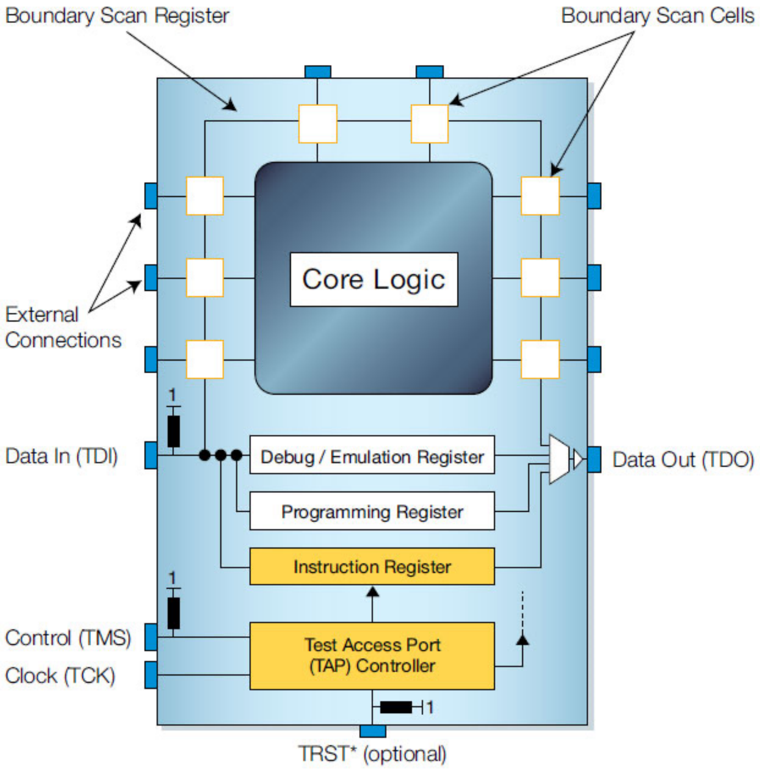
SPI is a synchronous serial communication protocol used for short-distance communication between a master and one or more slave devices. It uses separate lines for **data in**, **data out**, **clock**, and **chip select**, enabling high data rates and simple hardware. Communication is full-duplex.

I2C (Inter-Integrated Circuit)

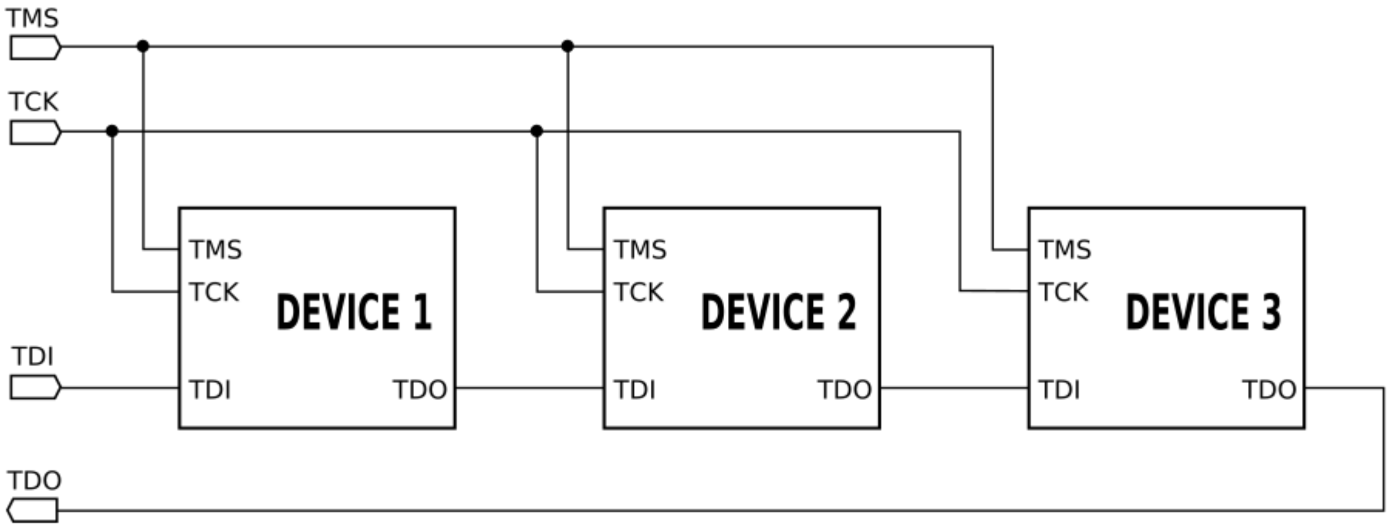
I2C is a synchronous, multi-master serial bus that uses only two lines: a **data line** and a **clock line**. Devices are addressed, allowing many peripherals to share the same bus. It is widely used for low-speed communication with sensors, EEPROMs, and configuration devices.



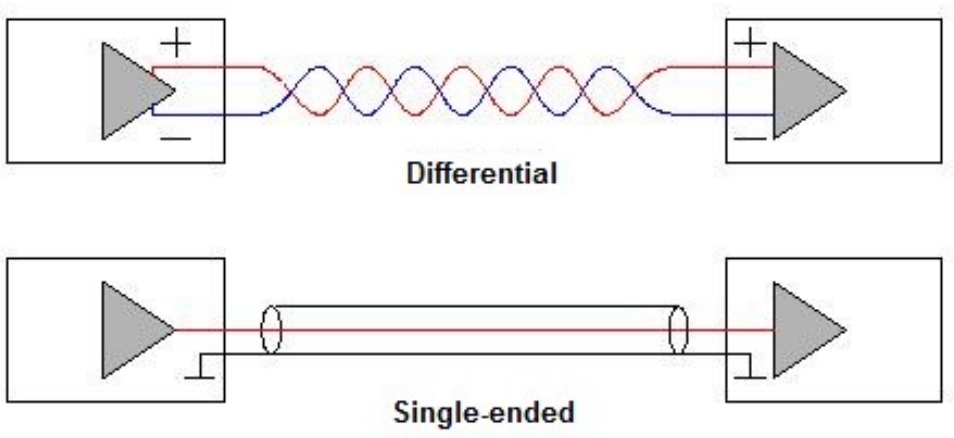
JTAG (Joint Test Action Group)



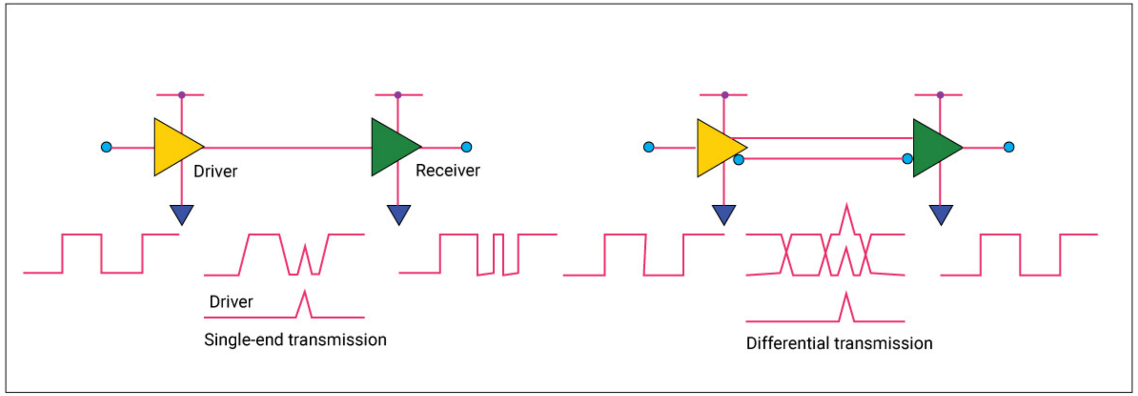
JTAG is a standard interface used for **testing, debugging, and programming electronic devices**. It provides access to internal registers and logic through a small set of dedicated pins. JTAG is commonly used for boundary-scan testing and in-system debugging of microcontrollers and FPGAs.



Single ended & Differential signaling

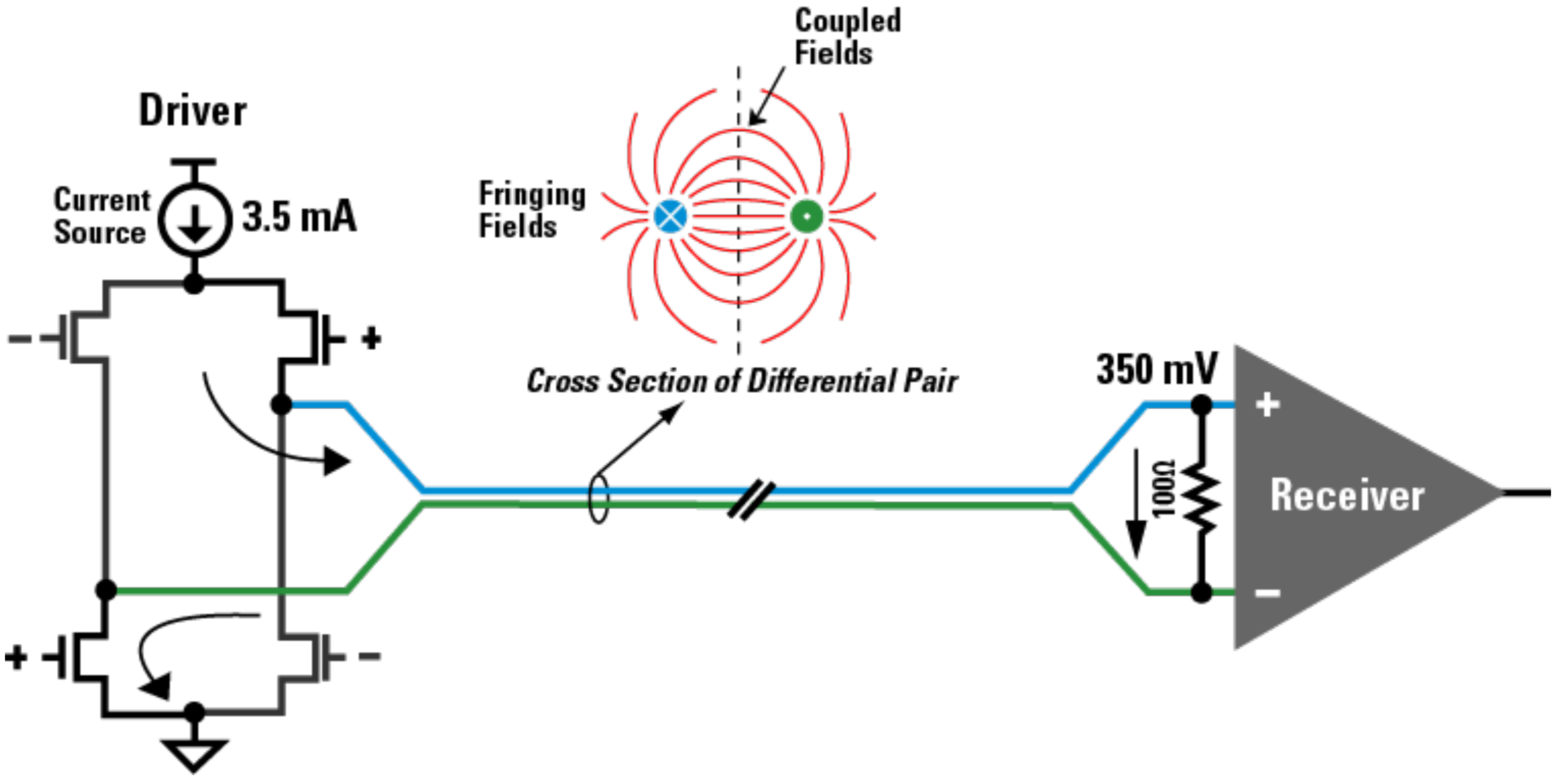


Single-ended signaling transmits information using one signal line referenced to a common ground. It is simple and low-cost but more susceptible to noise, ground offsets, and electromagnetic interference, especially at high speeds or long distances.

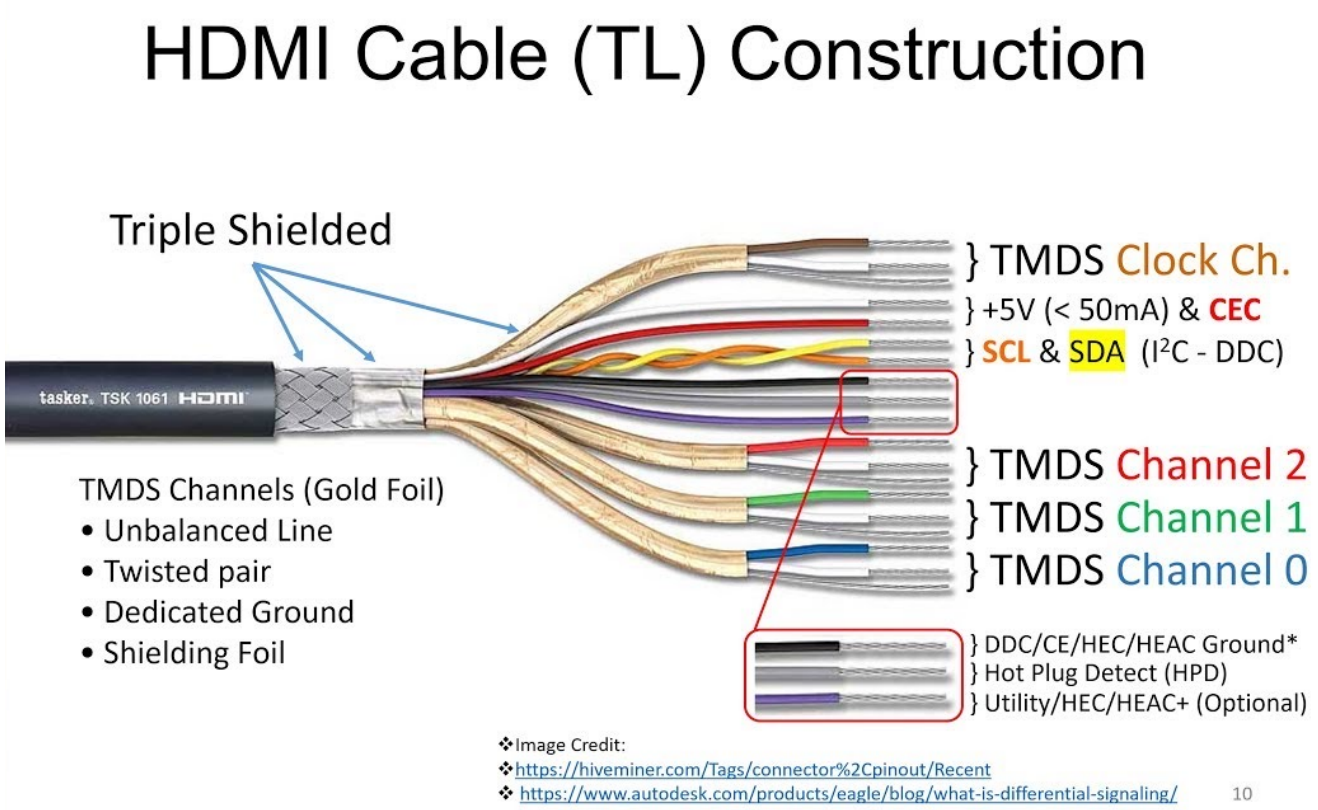


Differential signaling uses two complementary signal lines, and information is conveyed by the voltage difference between them.

LVDS (Low-voltage differential signaling)



TMDS (Transition-minimized differential signaling)



Signaling standards

SelectIO™ Interface DC Input and Output Levels

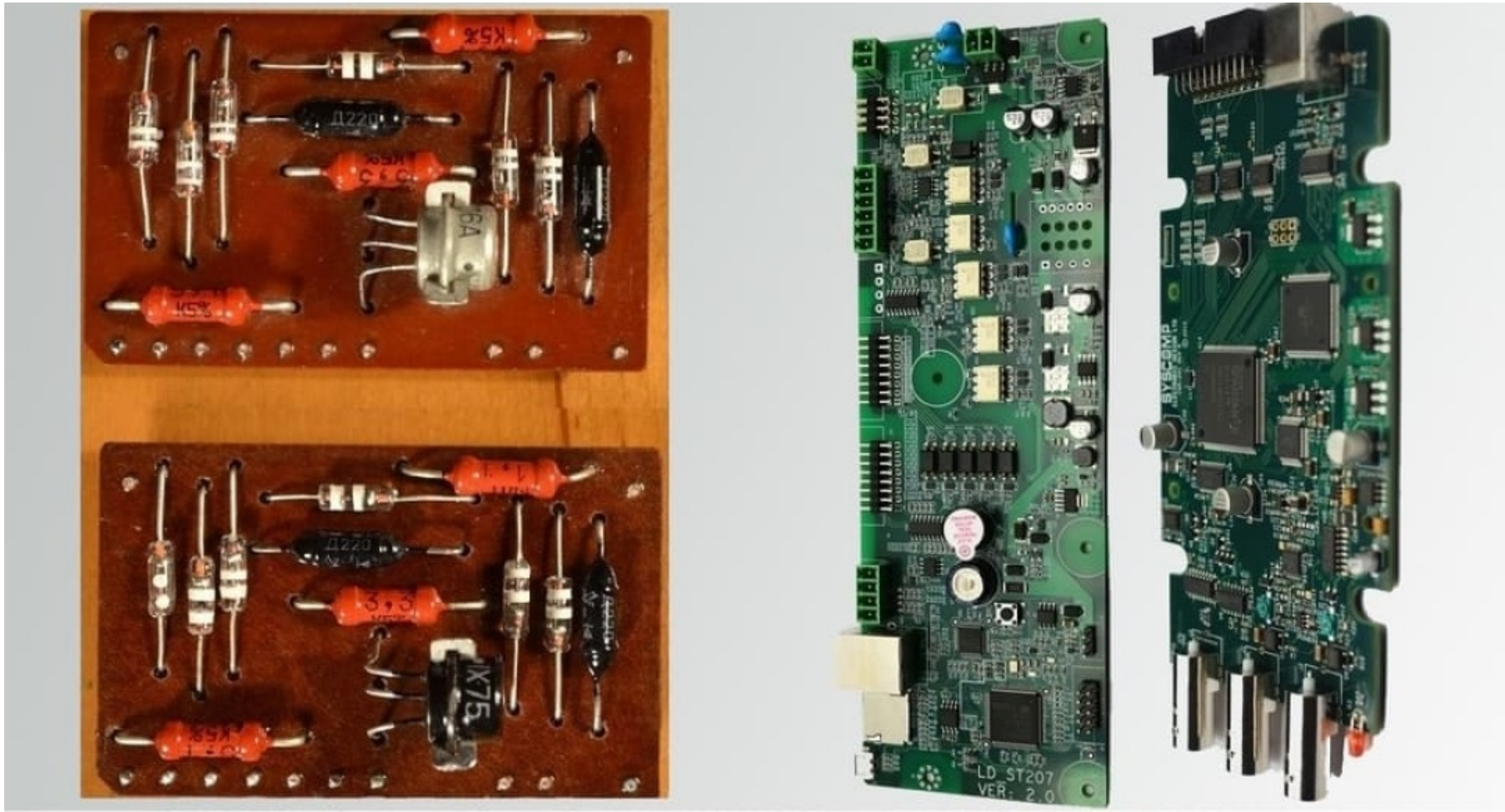
Table 7: **Recommended Operating Conditions for User I/Os Using Single-Ended Standards**

I/O Standard	V_{CCO} for Drivers[1]			V_{REF} for Inputs		
	V, Min	V, Nom	V, Max	V, Min	V, Nom	V, Max
LVTTL	3.0	3.3	3.45			
LVCMOS33	3.0	3.3	3.45			
LVCMOS25	2.3	2.5	2.7			
LVCMOS18	1.65	1.8	1.95			

Table 9: **Single-Ended I/O Standard DC Input and Output Levels**

I/O Standard	V_{IL}		V_{IH}		V_{OL}	V_{OH}	I_{OL}	I_{OH}
	V, Min	V, Max	V, Min	V, Max	V, Max	V, Min	mA	mA
LVTTL	–0.5	0.8	2.0	4.1	0.4	2.4	Note 2	Note 2
LVCMOS33	–0.5	0.8	2.0	4.1	0.4	$V_{CCO} - 0.4$	Note 2	Note 2
LVCMOS25	–0.5	0.7	1.7	4.1	0.4	$V_{CCO} - 0.4$	Note 2	Note 2
LVCMOS18	–0.5	0.38	0.8	4.1	0.45	$V_{CCO} - 0.45$	Note 2	Note 2

PCB development and history



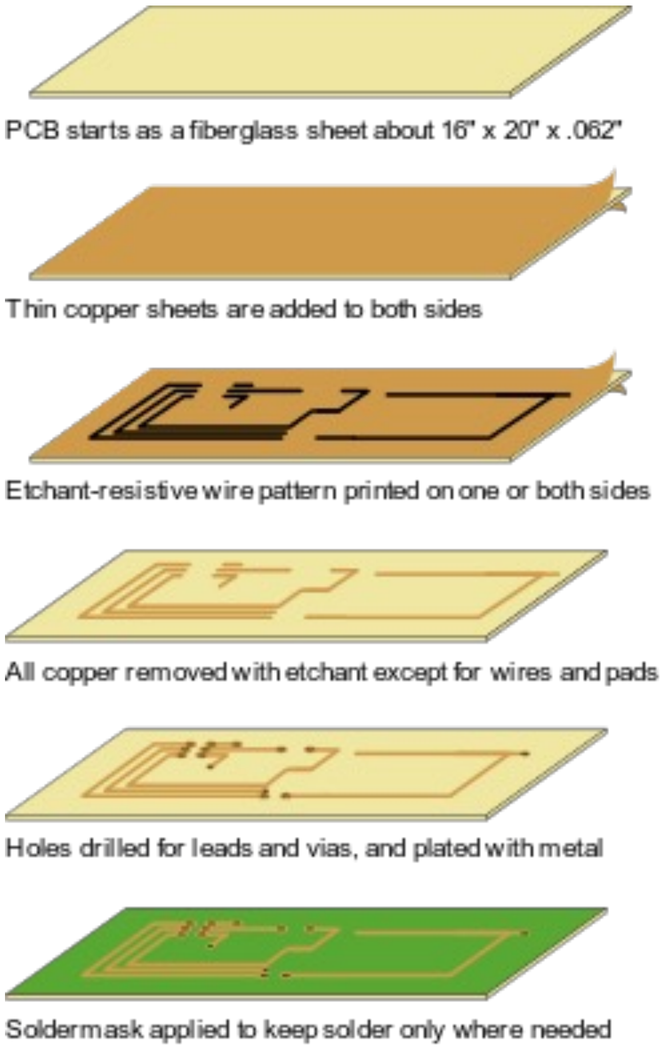
PCB in 1970s PCB Now

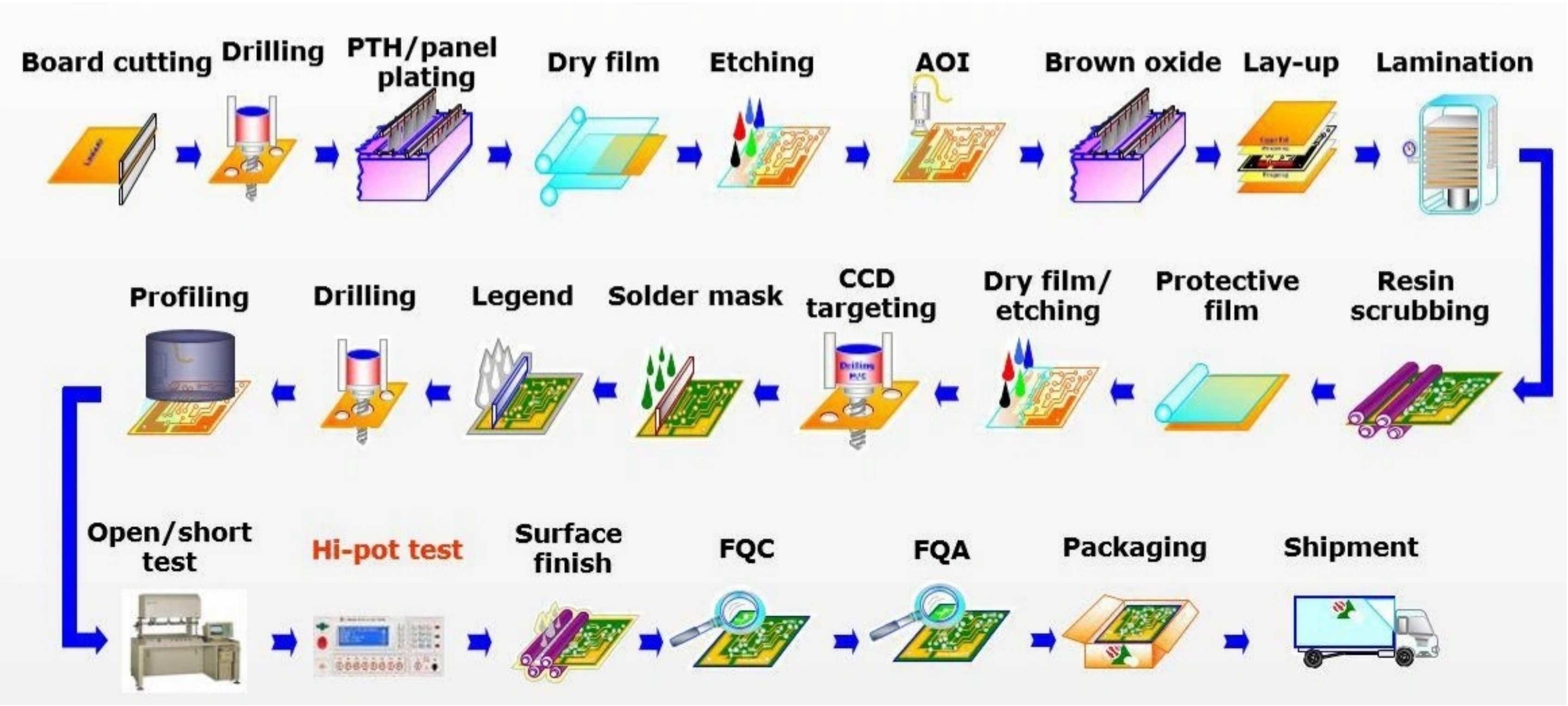
Why we need Printed Circuit Boards

An old Motorola television without PCB from 1948

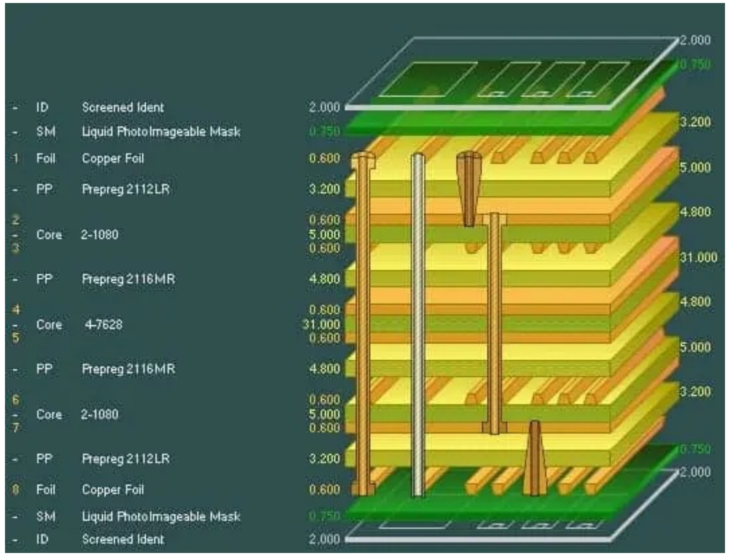
- Miniaturize devices
- Improve mass production
- Reduce cost
- Improve yeld
- Increase repeatability
- Reduced parasitics

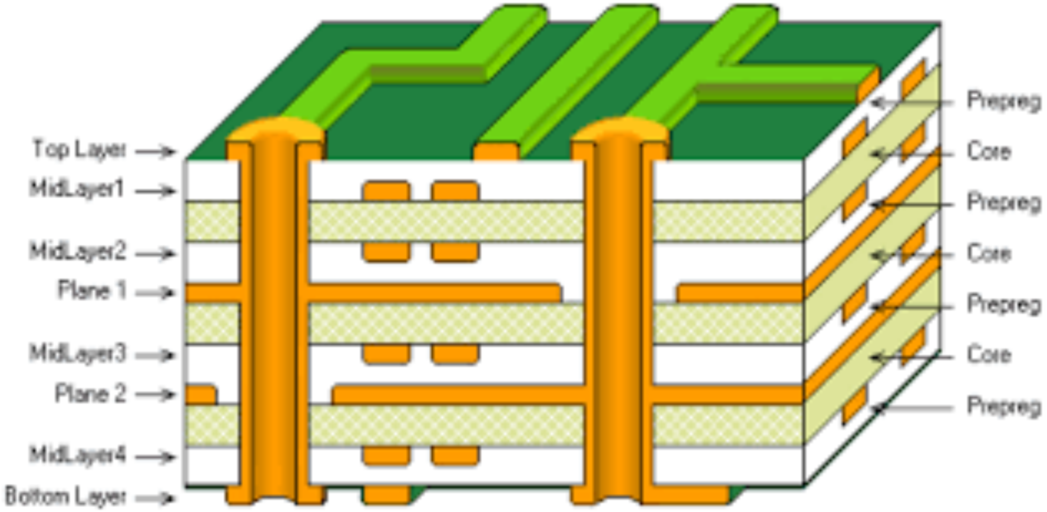
How a PCB is made?

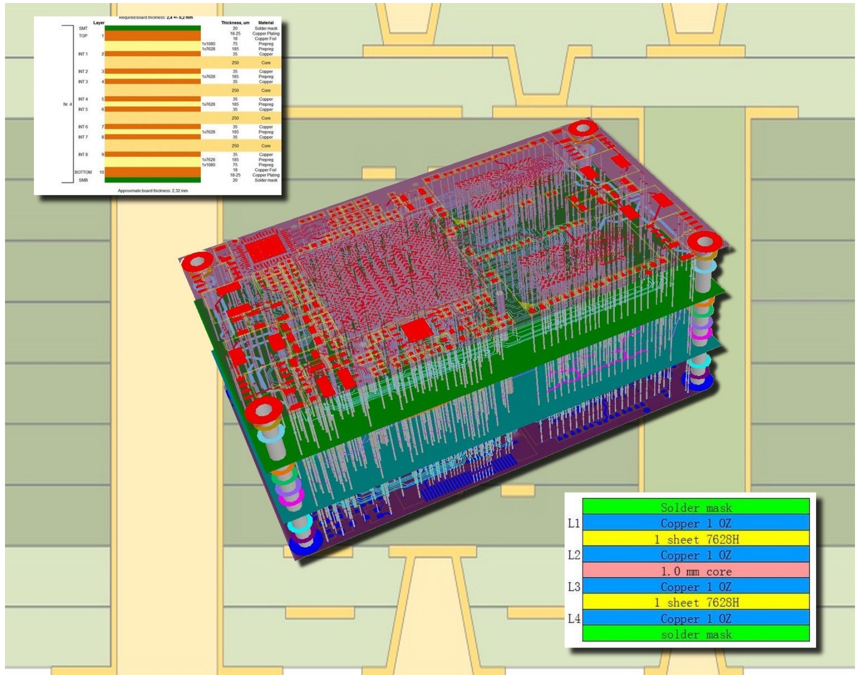


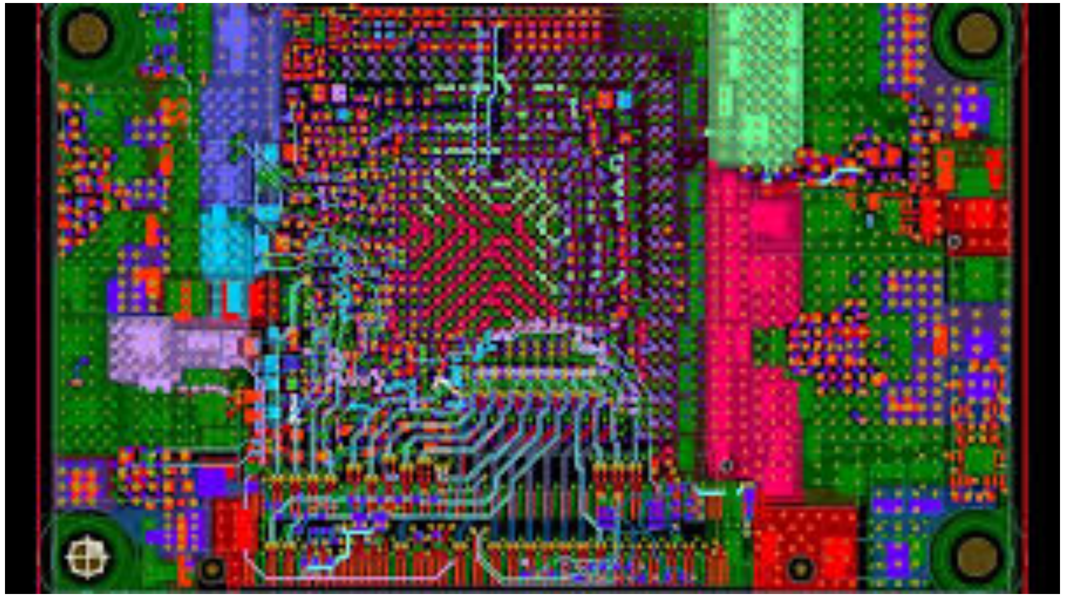


PCB stackup/1









PCB stackup/2

isola

185HR

High Reliability Epoxy Laminate and Prepreg

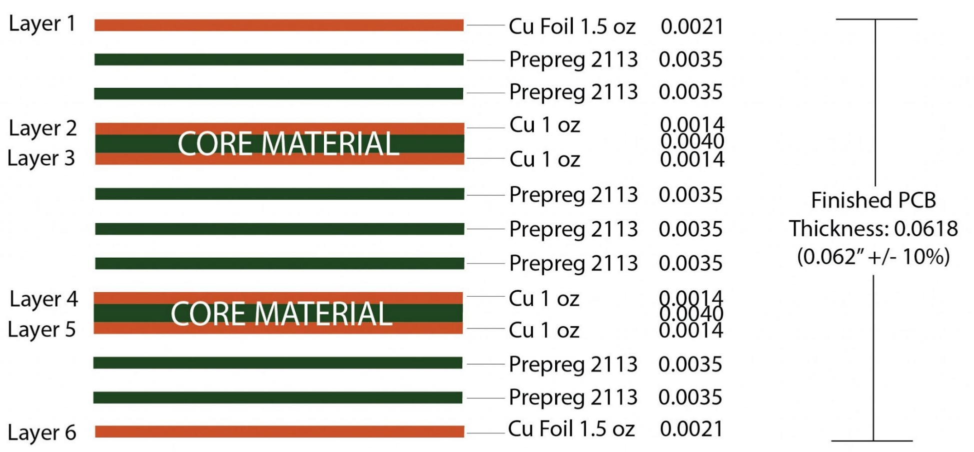
185HR laminate and prepreg materials are a proprietary, high-performance resin system with a Tg of 180°C for multilayer Printed Wiring Board (PWB) applications where maximum thermal performance and reliability are required.

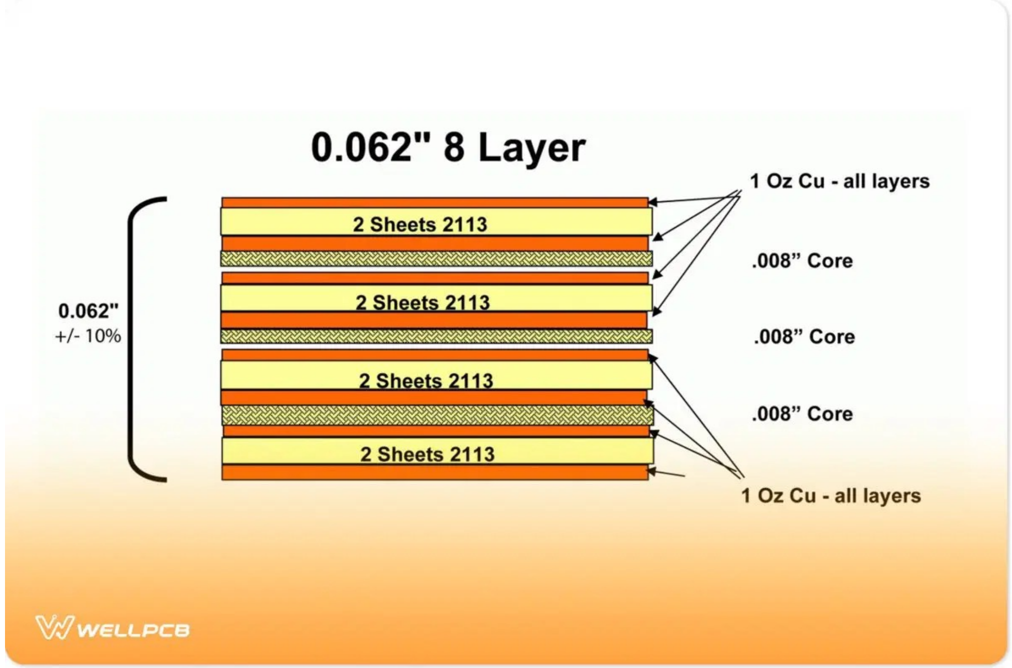
Constructions	Resin Content %	Standard/ Alternate	Thickness (inch)	Thickness (mm)	Dielectric Constant (DK)/ Dissipation Factor (DF)					
					100 MHz	500 MHz	1 GHz	2 GHz	5 GHz	10 GHz
1x1080	60%	Standard	0.0025	0.064	4.06 0.0160	3.98 0.0190	3.96 0.0200	3.93 0.0210	3.81 0.0250	3.80 0.0250
1x1067	73%	Alt/Spread	0.0025	0.064	3.84 0.0180	3.77 0.0210	3.74 0.0230	3.70 0.0240	3.55 0.0290	3.54 0.0290
1x1080	66%	Standard	0.0030	0.076	3.96 0.0170	3.89 0.0200	3.86 0.0220	3.82 0.0220	3.68 0.0270	3.68 0.0270
1x1086	61%	Alt/Spread	0.0030	0.076	4.04 0.0170	3.97 0.0190	3.94 0.0200	3.91 0.0210	3.78 0.0250	3.78 0.0250
1x3313	52%	Alternate	0.0035	0.089	4.20 0.0150	4.13 0.0170	4.11 0.0180	4.09 0.0190	3.97 0.0220	3.98 0.0220
1x2113	54%	Standard	0.0035	0.089	4.19 0.0160	4.10 0.0170	4.07 0.0190	4.04 0.0200	3.92 0.0230	3.92 0.0230
2x106	68%	Standard	0.0035	0.089	3.92 0.0140	3.88 0.0150	3.86 0.0180	3.84 0.0190	3.69 0.0190	3.69 0.0190
1x2116	48%	Standard	0.0040	0.102	4.28 0.0150	4.21 0.0160	4.20 0.0180	4.18 0.0180	4.05 0.0210	4.05 0.0210
1x3313	57%	Alternate	0.0040	0.102	4.11 0.0160	4.04 0.0180	4.03 0.0190	3.99 0.0200	3.86 0.0240	3.86 0.0240
2x106	72%	Standard	0.0040	0.102	3.86 0.0180	3.78 0.0210	3.76 0.0230	3.72 0.0240	3.57 0.0280	3.56 0.0290
1x2116	53%	Standard	0.0045	0.114	4.18 0.0150	4.11 0.0170	4.10 0.0190	4.07 0.0190	3.94 0.0230	3.94 0.0230
1x106 / 1x1080	65%	Alternate	0.0045	0.114	3.97 0.0170	3.90 0.0200	3.87 0.0210	3.84 0.0220	3.70 0.0260	3.70 0.0260
1x2116	57%	Standard	0.0050	0.127	4.11 0.0160	4.04 0.0180	4.03 0.0190	3.99 0.0200	3.86 0.0240	3.86 0.0240
2x1080	60%	Standard	0.0050	0.127	4.06 0.0160	3.98 0.0190	3.96 0.0200	3.93 0.0210	3.81 0.0250	3.80 0.0250
2x1067	73%	Alt/Spread	0.0050	0.127	3.84 0.0180	3.77 0.0210	3.74 0.0230	3.70 0.0240	3.55 0.0290	3.54 0.0290
2x1080	63%	Alternate	0.0055	0.140	4.00 0.0170	3.93 0.0190	3.91 0.0210	3.87 0.0220	3.74 0.0260	3.74 0.0260
1x106 / 1x2113	60%	Alternate	0.0055	0.140	4.06 0.0160	3.98 0.0190	3.96 0.0200	3.93 0.0210	3.81 0.0250	3.80 0.0250
1x1652	52%	Standard	0.0060	0.152	4.20 0.0150	4.13 0.0170	4.11 0.0180	4.09 0.0190	3.97 0.0220	3.98 0.0220
2x1080	65%	Standard	0.0060	0.152	3.97 0.0170	3.90 0.0200	3.87 0.0210	3.84 0.0220	3.70 0.0260	3.70 0.0260

Property		Typical Value	Units	Test Method
			Metric (English)	IPC-TM-650 (or as noted)
Test data generated from rigid laminate		50	%	2.3.16.2
Glass Transition Temperature (Tg) by DSC		180	°C	2.4.25C
Glass Transition Temperature (Tg) by DMA		185	°C	2.4.24.4
Decomposition Temperature (Td) by TGA @ 5% weight loss		340	°C	2.4.24.6
Time to Delaminate by TMA (Copper removed)	A. T260 B. T288	60 >15	Minutes	2.4.24.1
Z-Axis CTE	A. Pre-Tg B. Post-Tg C. 50 to 260°C, (Total Expansion)	40 220 2.7	ppm/°C ppm/°C %	2.4.24C
X/Y-Axis CTE	Pre-Tg	13/14	ppm/°C	2.4.24C
Thermal Conductivity		0.4	W/m·K	ASTM E1952
Thermal Stress 10 sec @ 288°C (550.4°F)	A. Unetched B. Etched	Pass	Pass Visual	2.4.13.1
Dk, Permittivity	A. @ 100 MHz B. @ 1 GHz C. @ 2 GHz D. @ 5 GHz E. @ 10 GHz	4.13 4.04 4.01 3.88 3.88	—	2.5.5.3 Bereskin Stripline Bereskin Stripline Bereskin Stripline Bereskin Stripline
Df, Loss Tangent	A. @ 100 MHz B. @ 1 GHz C. @ 2 GHz D. @ 5 GHz E. @ 10 GHz	0.0158 0.0192 0.0200 0.0235 0.0236	—	2.5.5.3 Bereskin Stripline Bereskin Stripline Bereskin Stripline Bereskin Stripline
Volume Resistivity	A. C-96/35/90 B. After moisture resistance C. At elevated temperature	— 3.0×10^8 7.0×10^8	MΩ-cm	2.5.17.1
Surface Resistivity	A. C-96/35/90 B. After moisture resistance C. At elevated temperature	— 3.0×10^6 2.0×10^8	MΩ	2.5.17.1
Dielectric Breakdown		>50	kV	2.5.6B
Arc Resistance		115	Seconds	2.5.1B
Electric Strength (Laminate & laminated prepreg)		54 (1350)	kV/mm (V/mil)	2.5.6.2A
Comparative Tracking Index (CTI)		3 (175-249)	Class (Volts)	UL 746A ASTM D3638
Peel Strength	A. Low profile copper foil and very low profile copper foil all copper foil >17 µm [0.669 mil] B. Standard profile copper 1. After thermal stress 2. At 125°C (257°F) 3. After process solutions	>0.79 (>4.5) >0.79 (>4.5) >0.79 (>4.5) >0.79 (>4.5)	N/mm (lb/inch)	2.4.8C 2.4.8.2A 2.4.8.3 2.4.8.3
Flexural Strength	A. Length direction B. Cross direction	669 (97.1) 373 (54.1)	MPa (kpsi)	2.4.4B

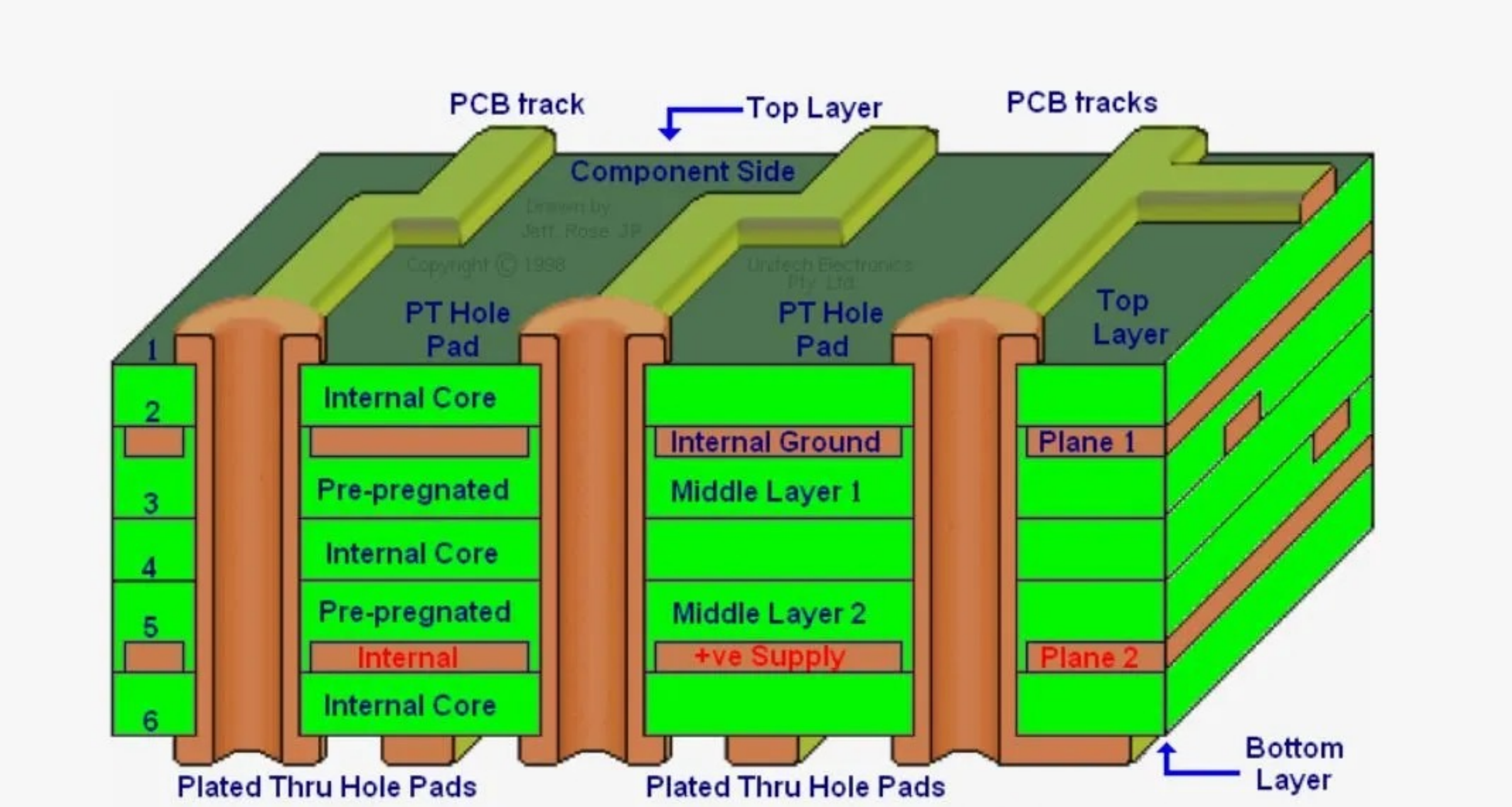
Nominal dimensions

Copper Weight	Copper Thickness
1 oz	1.37 mils/0.0348mm
2 oz	2.74 mils/0.0696mm
3 oz	4.11 mils/0.1044mm
4 oz	5.48 mils/0.1392mm

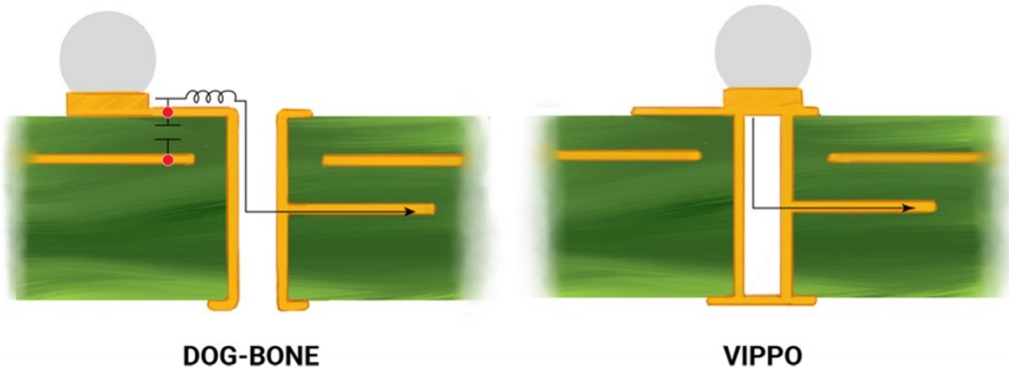


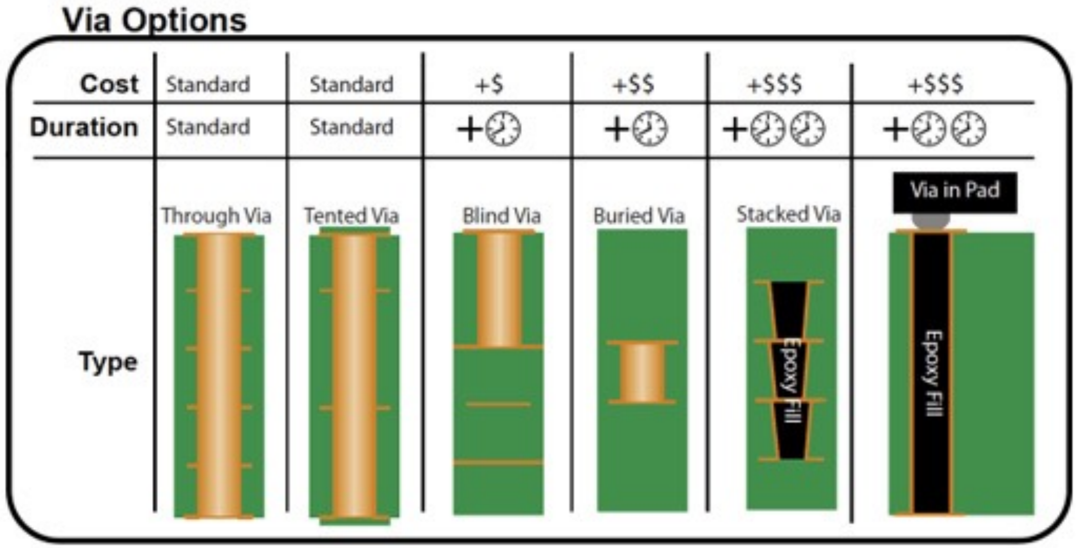


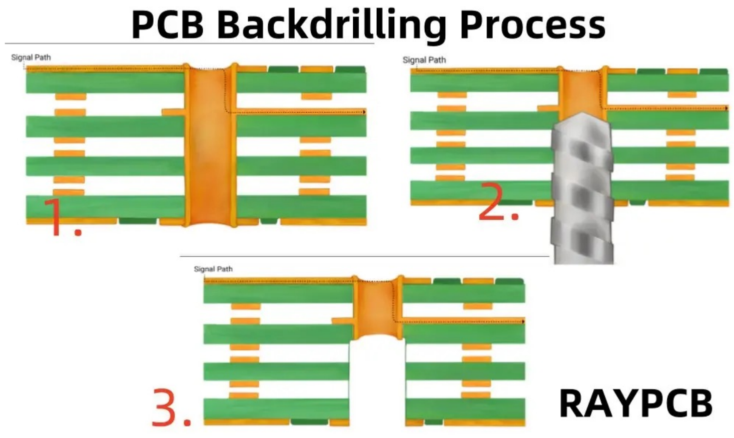
Some nomenclature

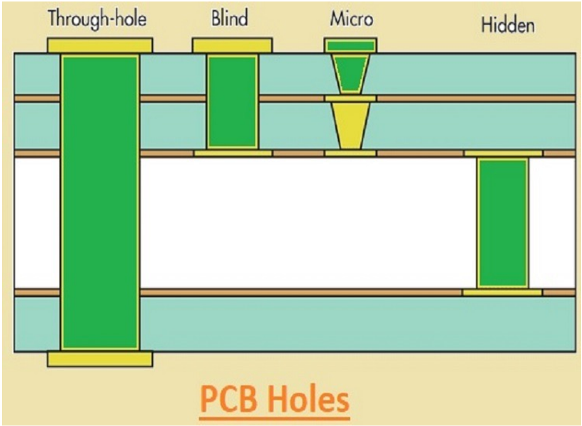


VIAs

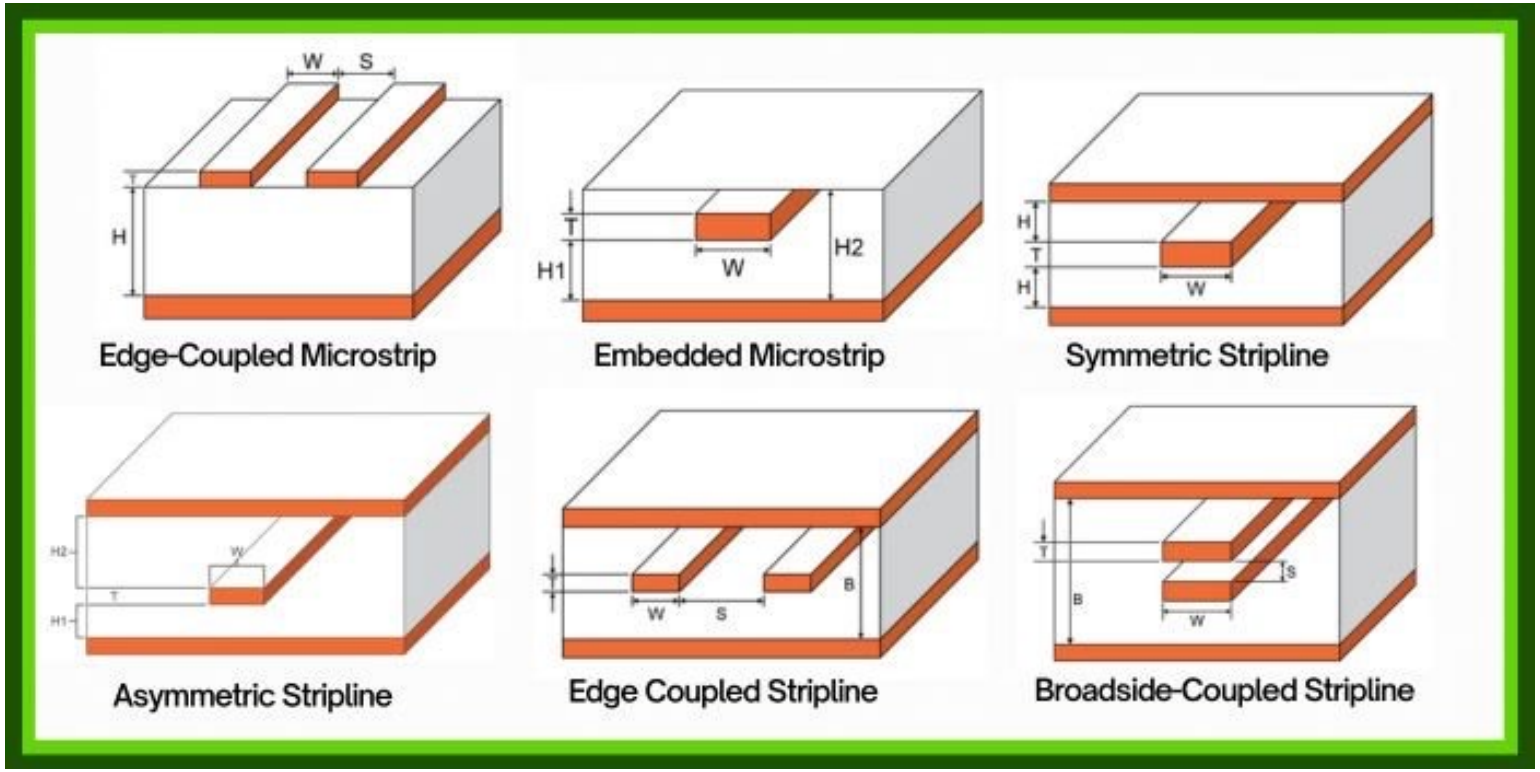

DOG-BONE
VIPPO


Via Options
Cost
Standard
Standard
+$
+$$
+$$$
+$$$
Duration
Standard
Standard
Type
Through Via
Tented Via
Blind Via
Buried Via
Stacked Via
Via in Pad
Epoxy Fill
Epoxy Fill


PCB Backdrilling Process
1.
2.
3.
RAYPCB


Through-hole
Blind
Micro
Hidden
PCB Holes

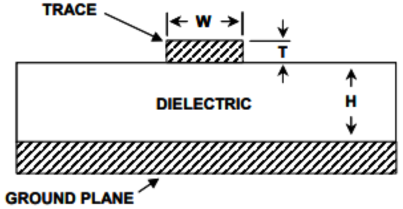
Tracks



W
S
H
Edge-Coupled Microstrip
T
H1
W
H2
Embedded Microstrip
H
T
H
W
Symmetric Stripline
Asymmetric Stripline
W
S
B
Edge Coupled Stripline
S
B
W
Broadside-Coupled Stripline

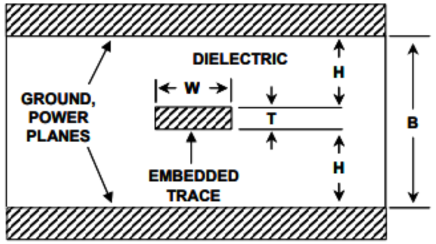
Track impedance

$$Z = \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln\left(\frac{5.98H}{0.8W + T}\right)$$



“In electrical engineering, impedance is the opposition to alternating current presented by the combined effect of resistance and reactance in a circuit.”

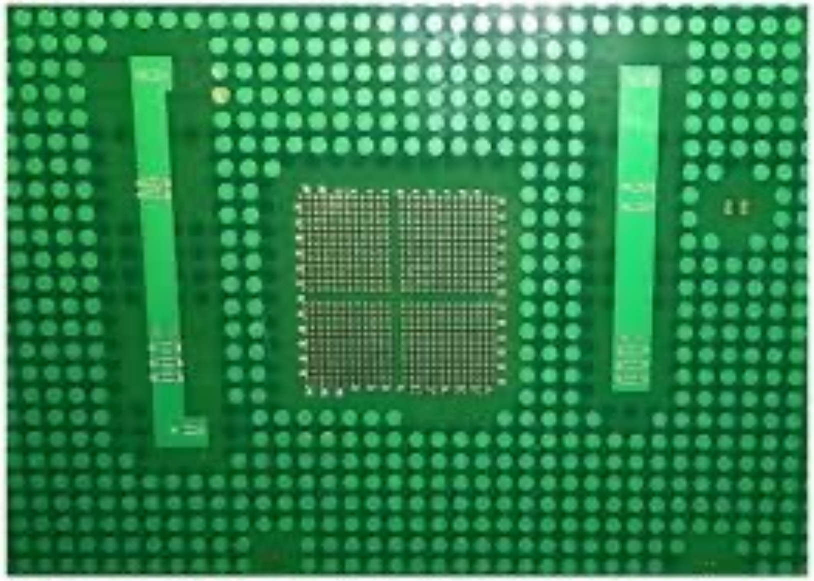
$$Z = \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{1.9B}{0.8W + T}\right)$$

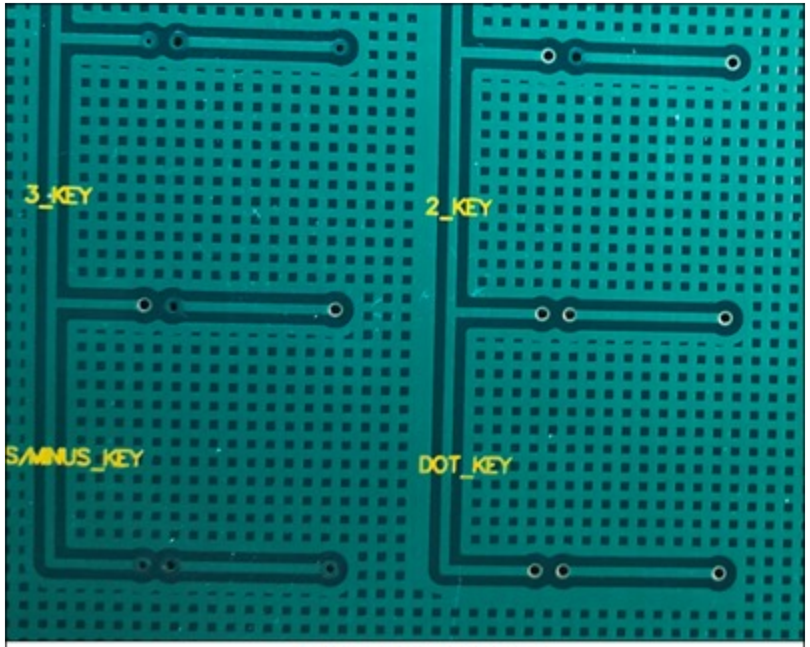


The characteristic impedance of a line is the impedance seen by a wave front traveling down the line

The trace width, dielectric thickness, and dielectric material are the only factors that set the impedance.

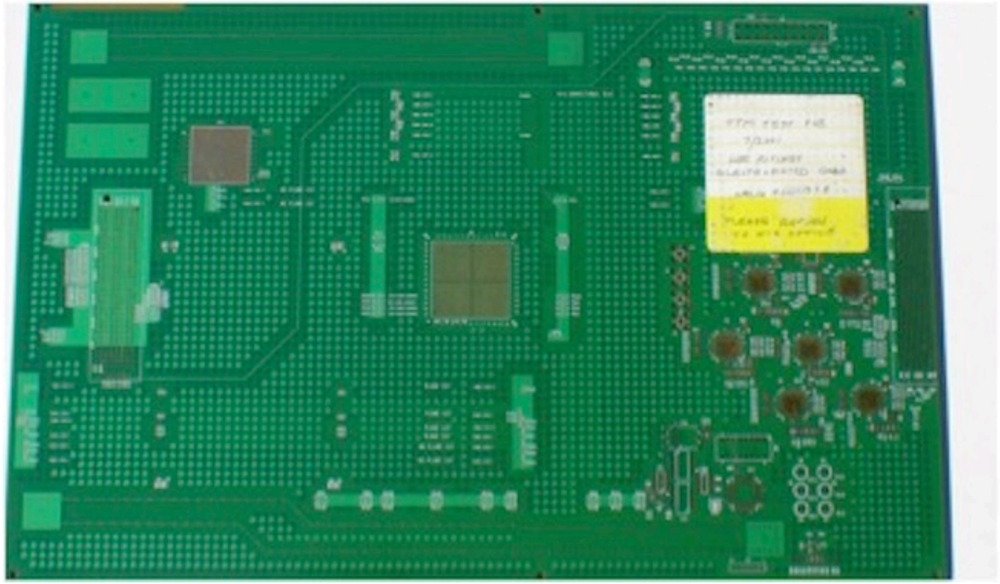
Copper thieving





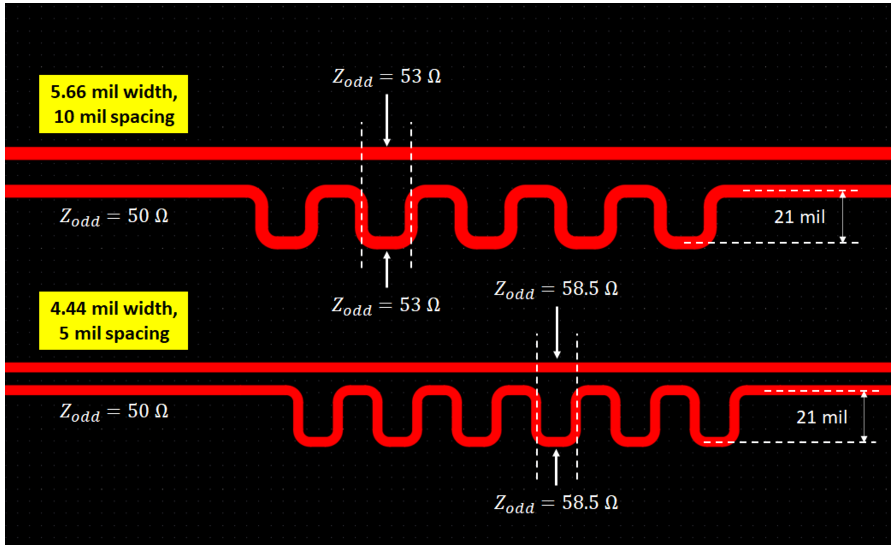
Copper Thieving Pattern

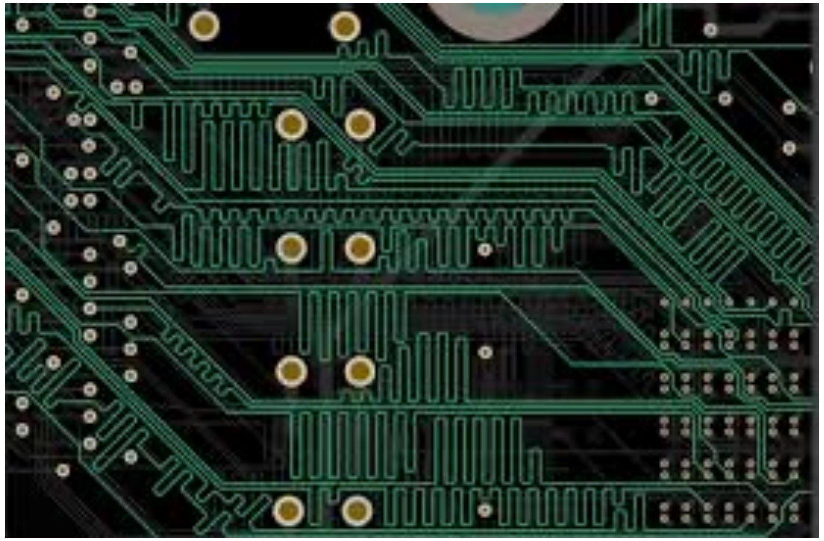
Used to even out the copper distribution across a copper layer. The added copper isn't connected to any nets on the board and should not affect the functionality.

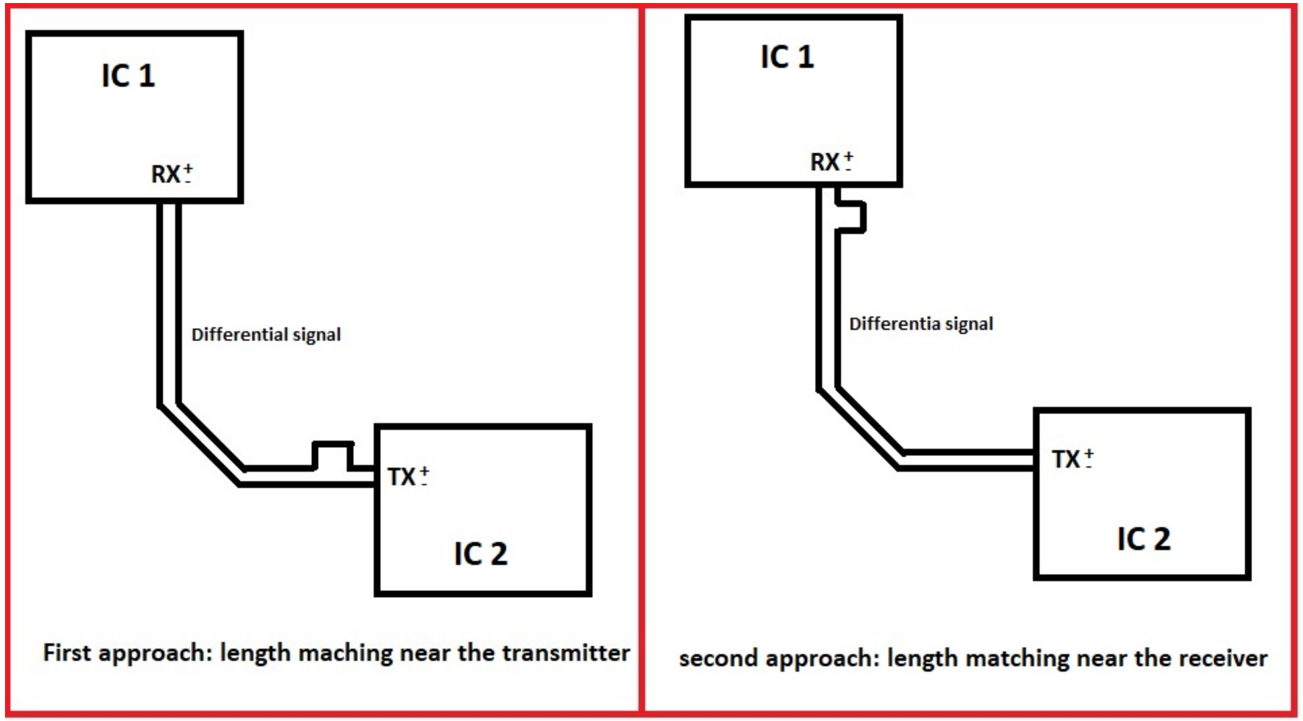


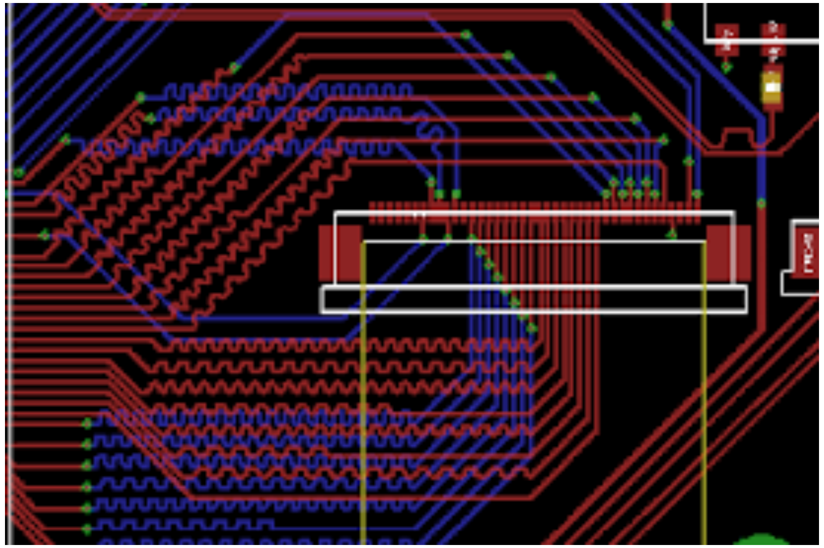
Care will be taken to ensure thieving will not interfere with any defined controlled impedance or RF signal lines the design may contain.

Length tuning

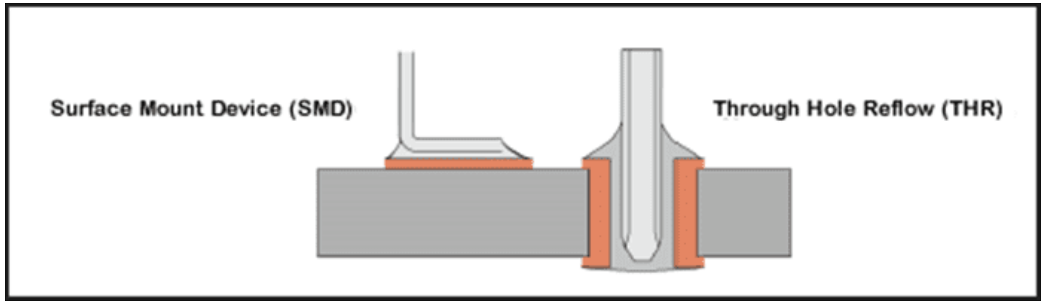








Components, SMD and THR



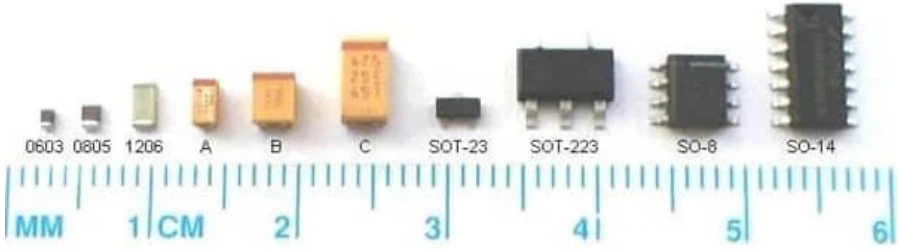
Surface Mount Device

- Better for automatic assembly
- Generally lower parasitics, less exposed metal, better for high frequency applications
- Can have a much smaller footprint

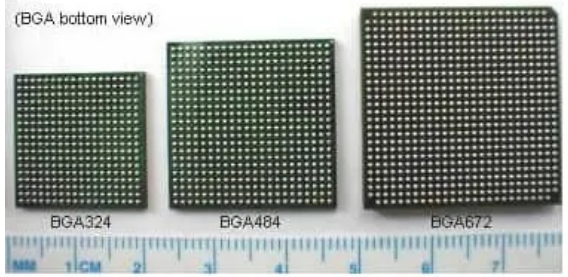
Through Hole Reflow

- Connection on all PCB layers
- Usable with breadboard and easier to hand solder
- Stronger mechanical connection

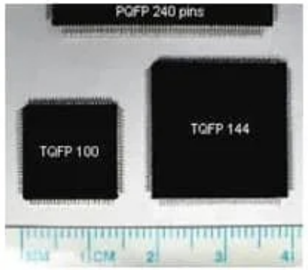
Footprint example for SMF and THR



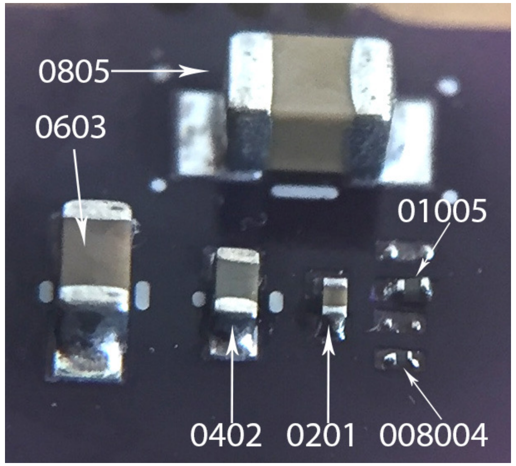
SMD electronic component size

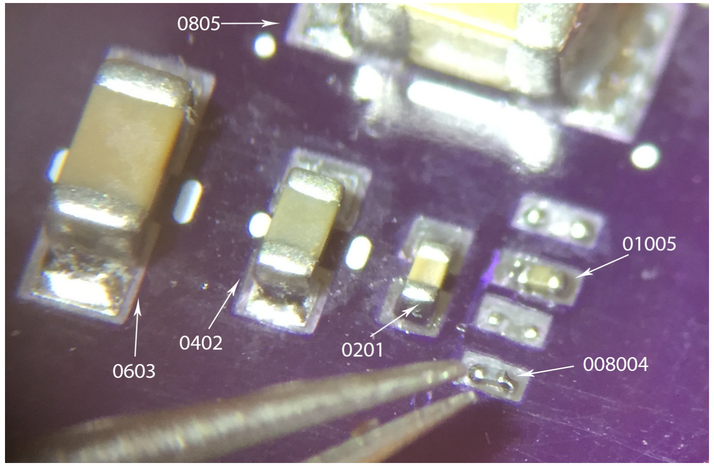


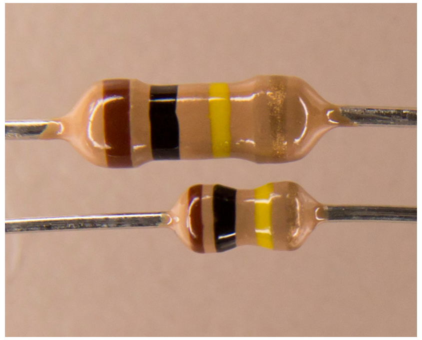
The Size of SMD BGA

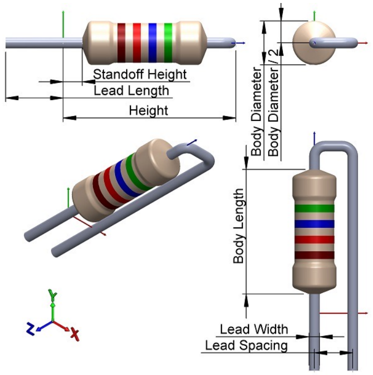


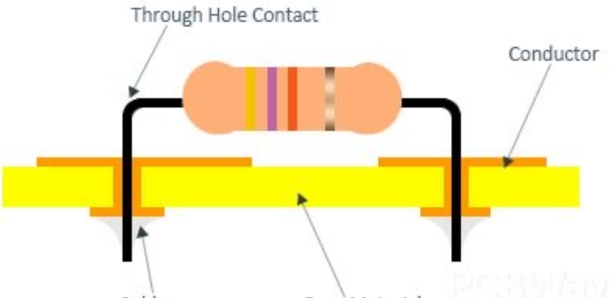
The Size of QFP

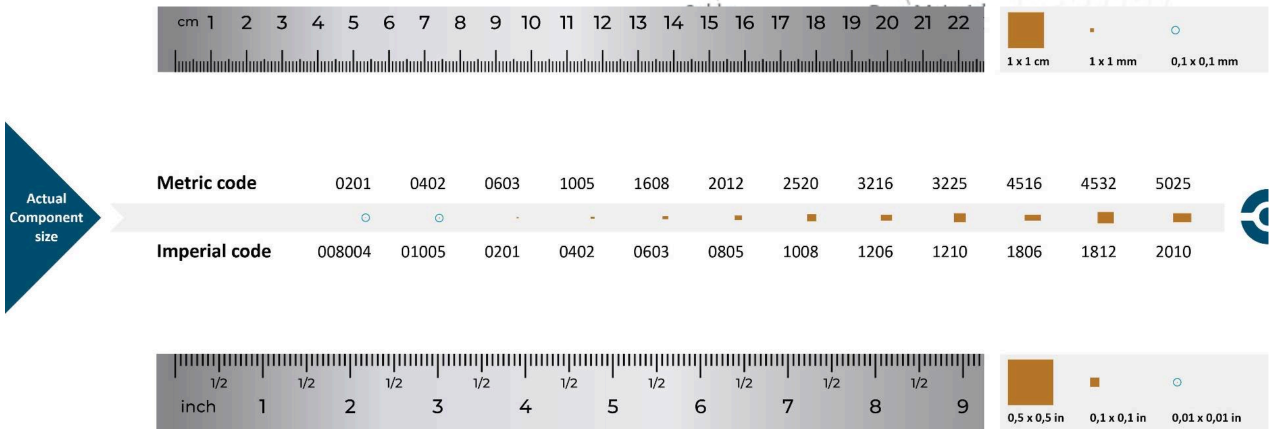




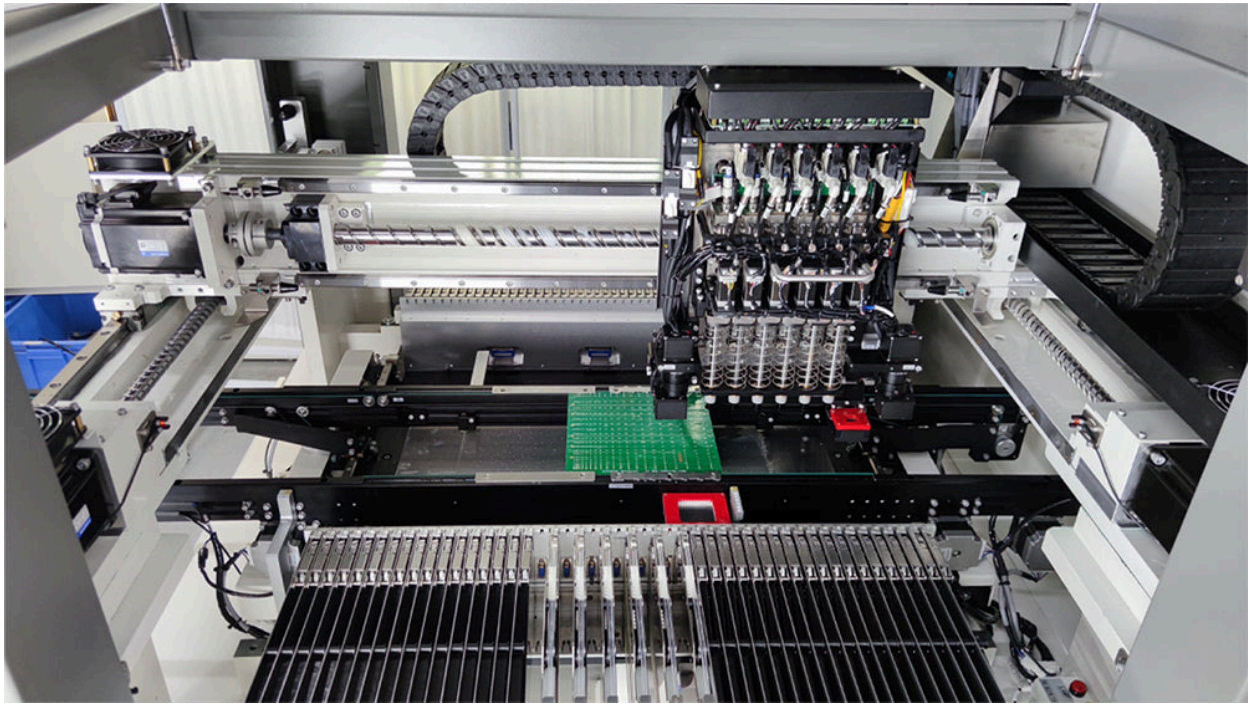


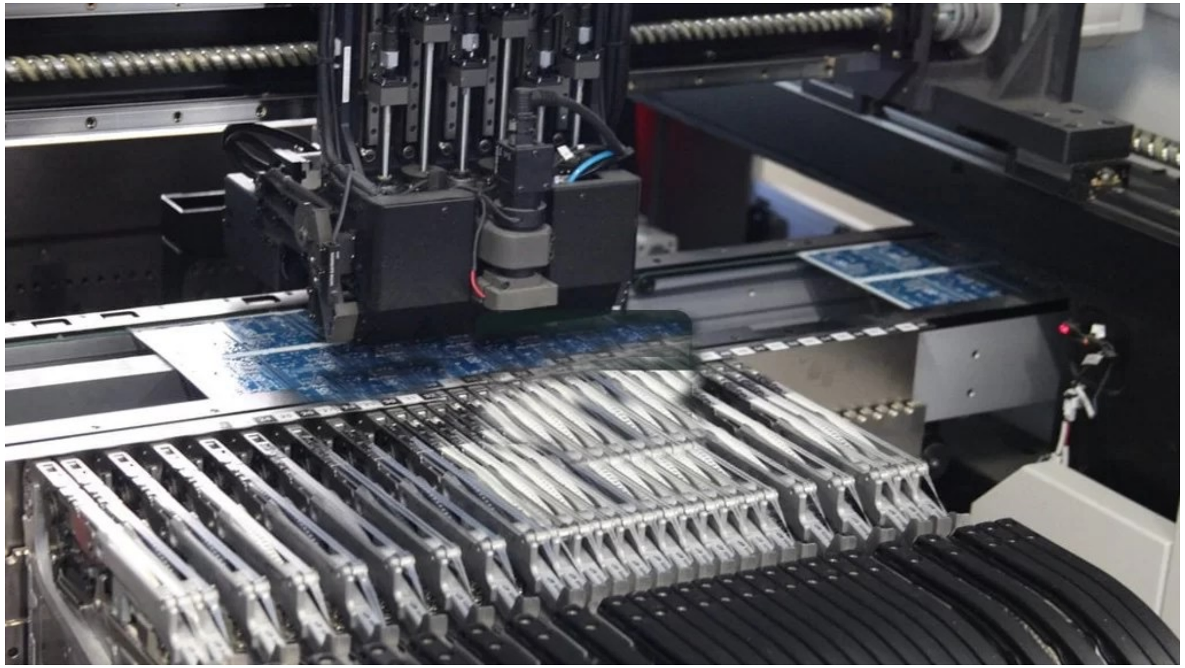




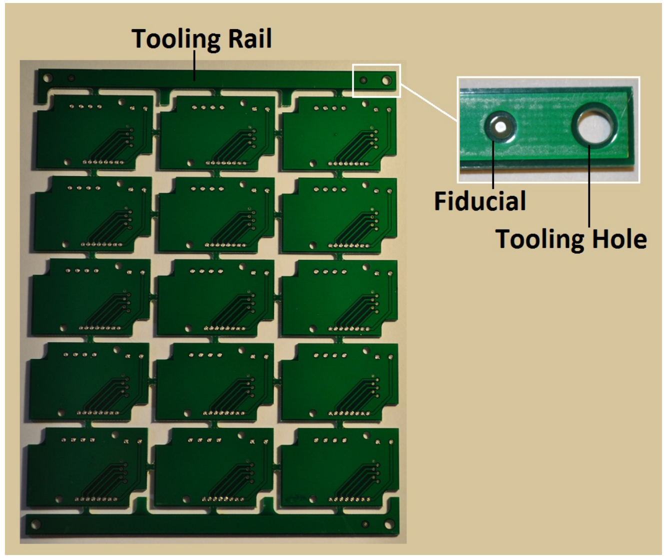


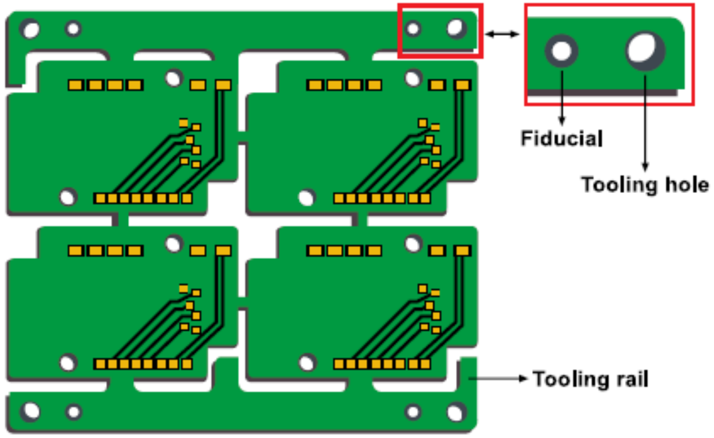
Pick and Place machine

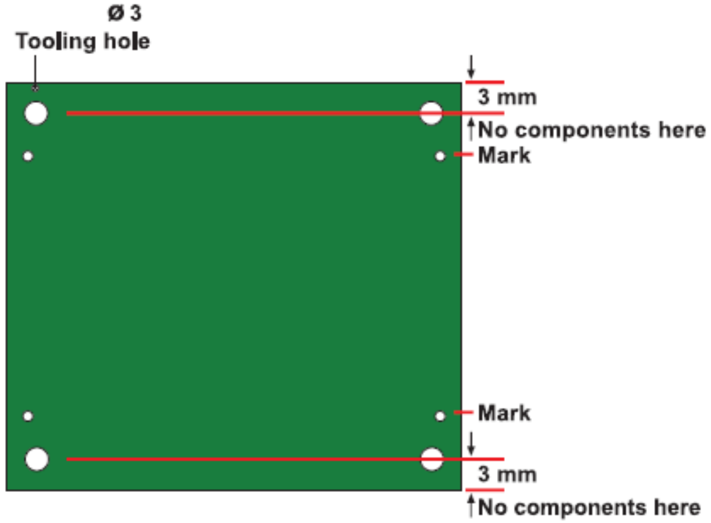


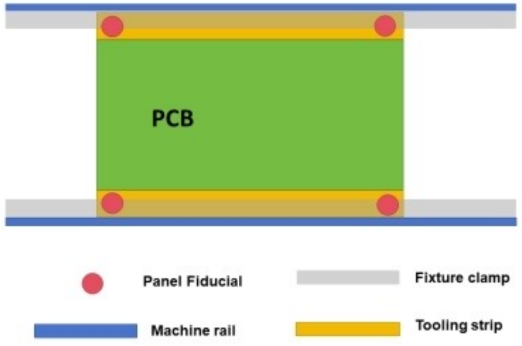


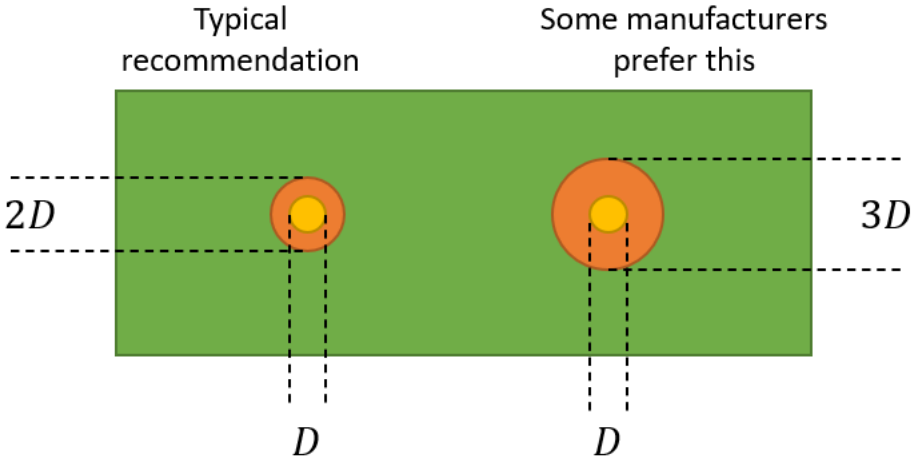
Fiducial and tooling holes



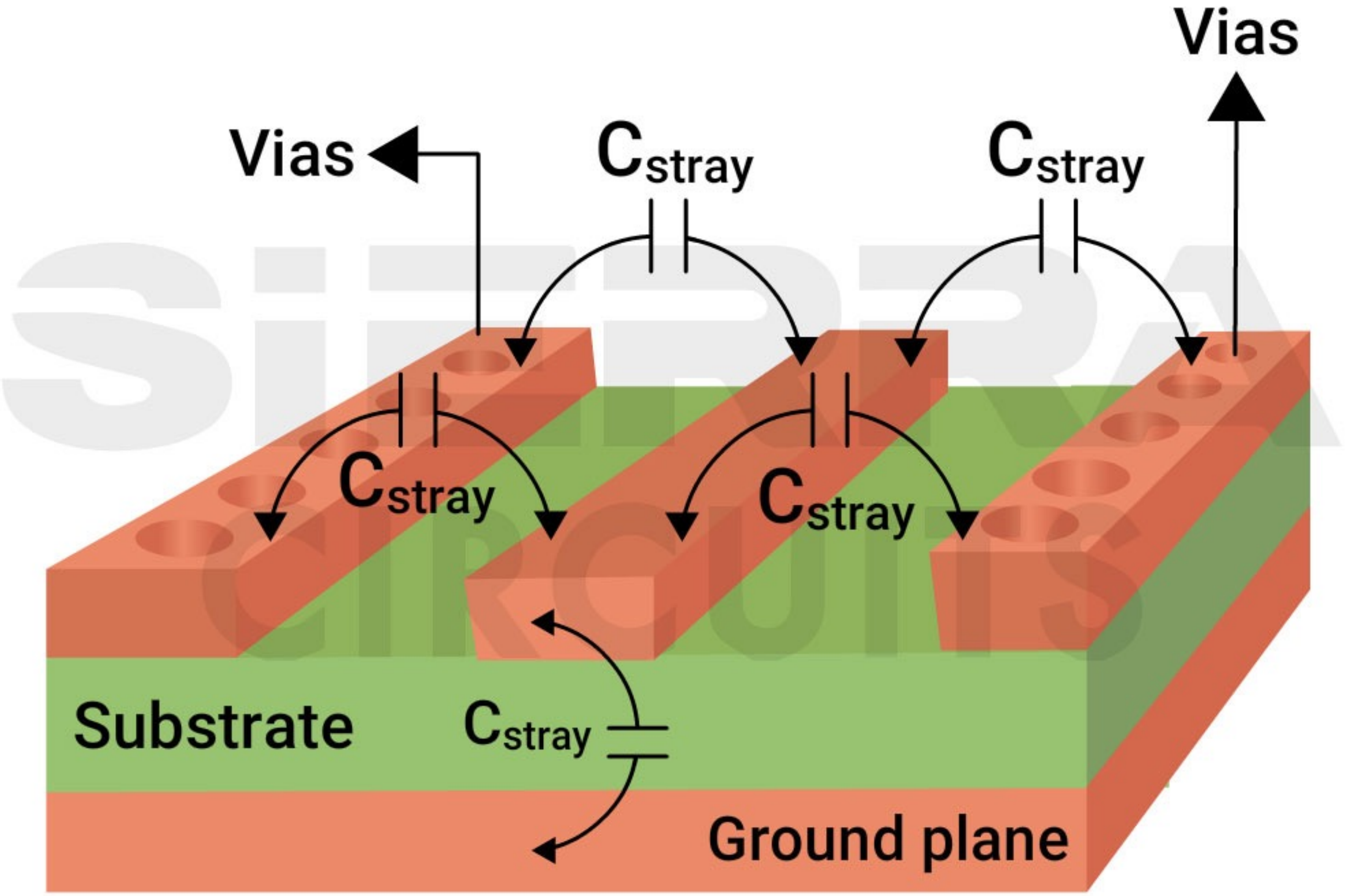




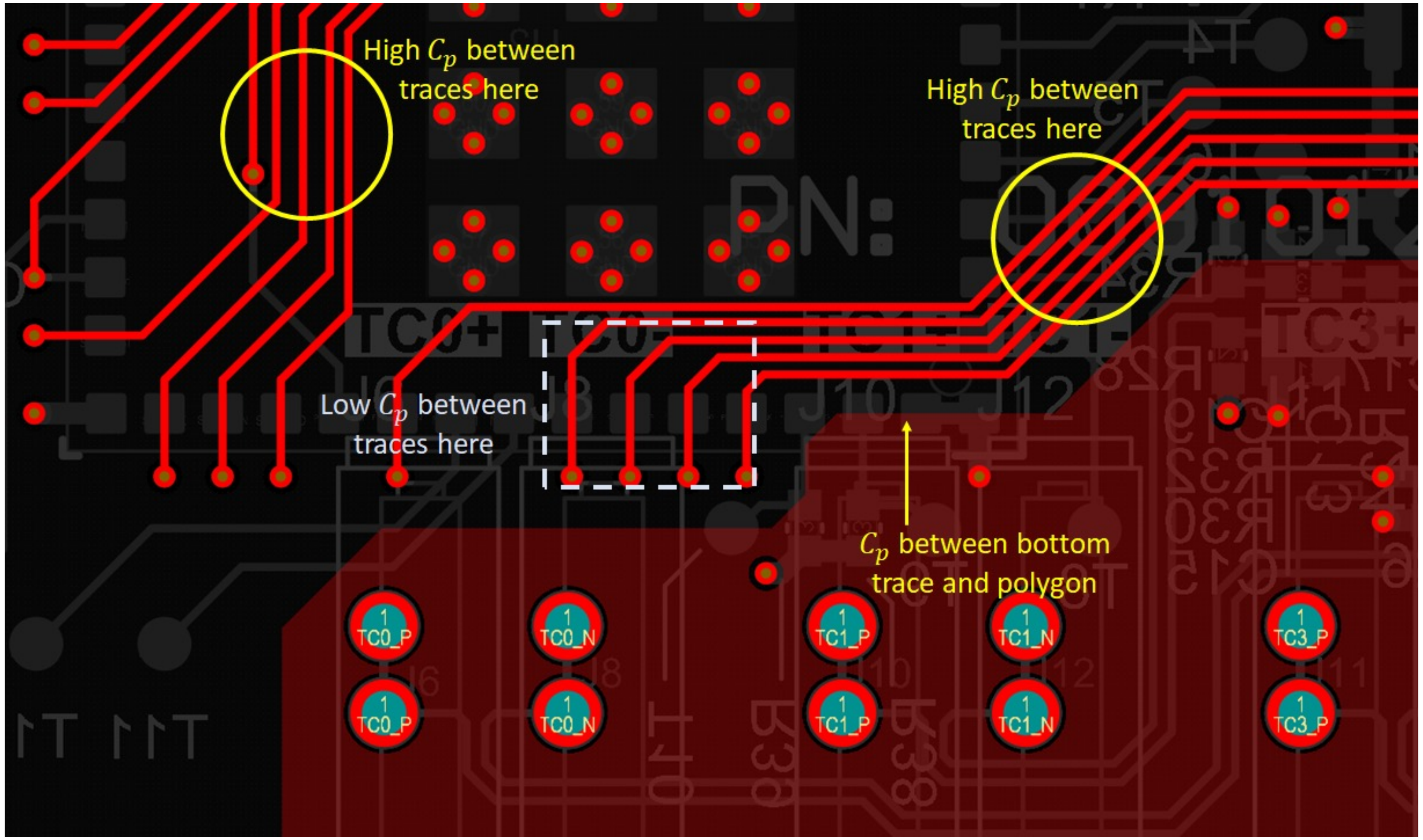


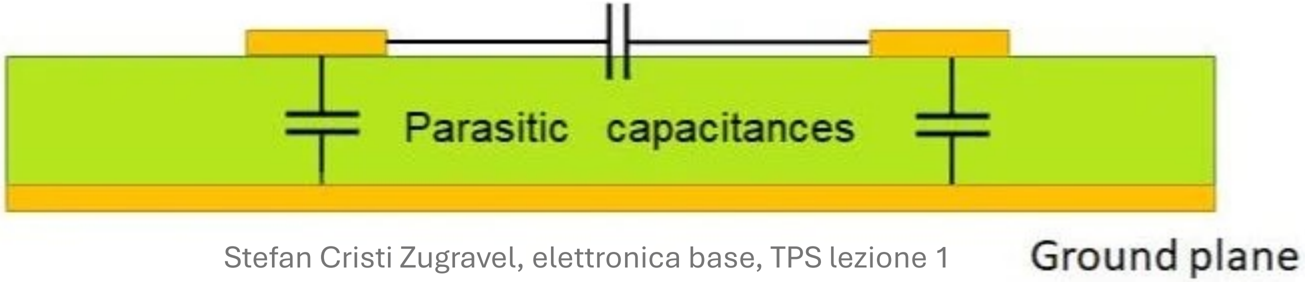


PCB parasitics/1

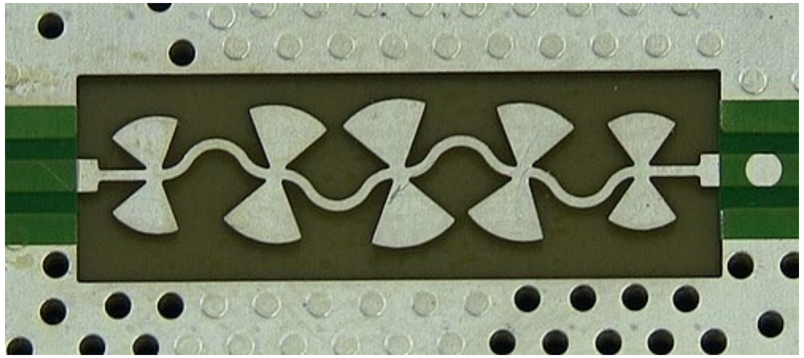


PCB parasitics/2

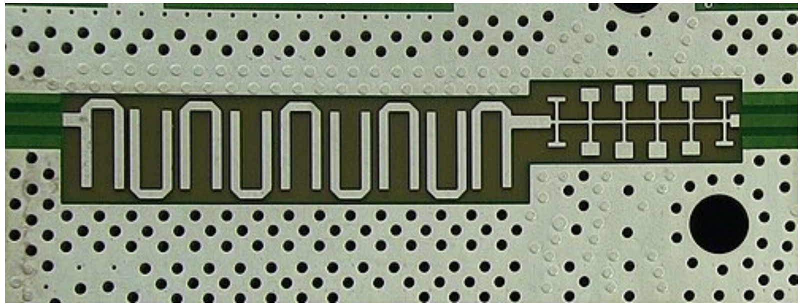




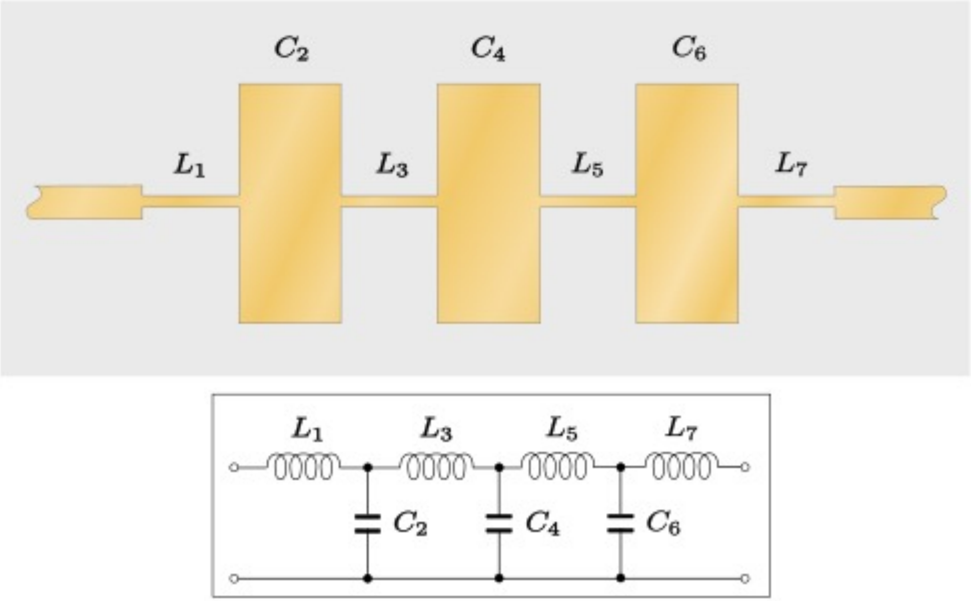
Distributed-element filter



A microstrip low pass filter implemented with bowtie stubs inside a 20 GHz Agilent N9344C spectrum analyzer

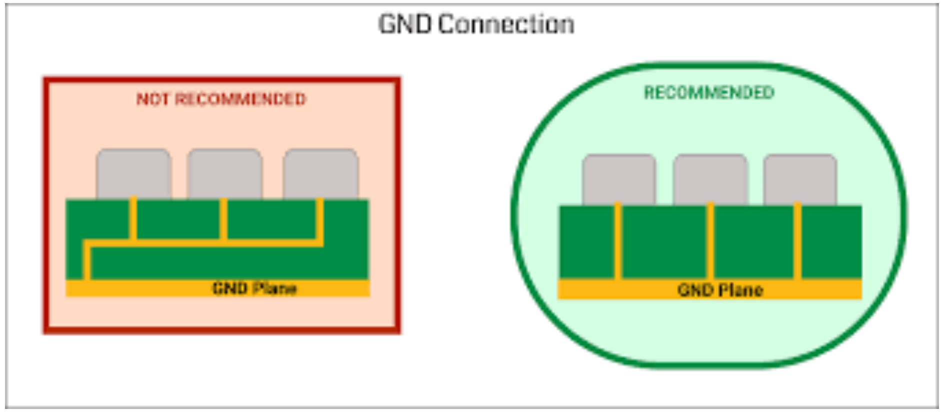


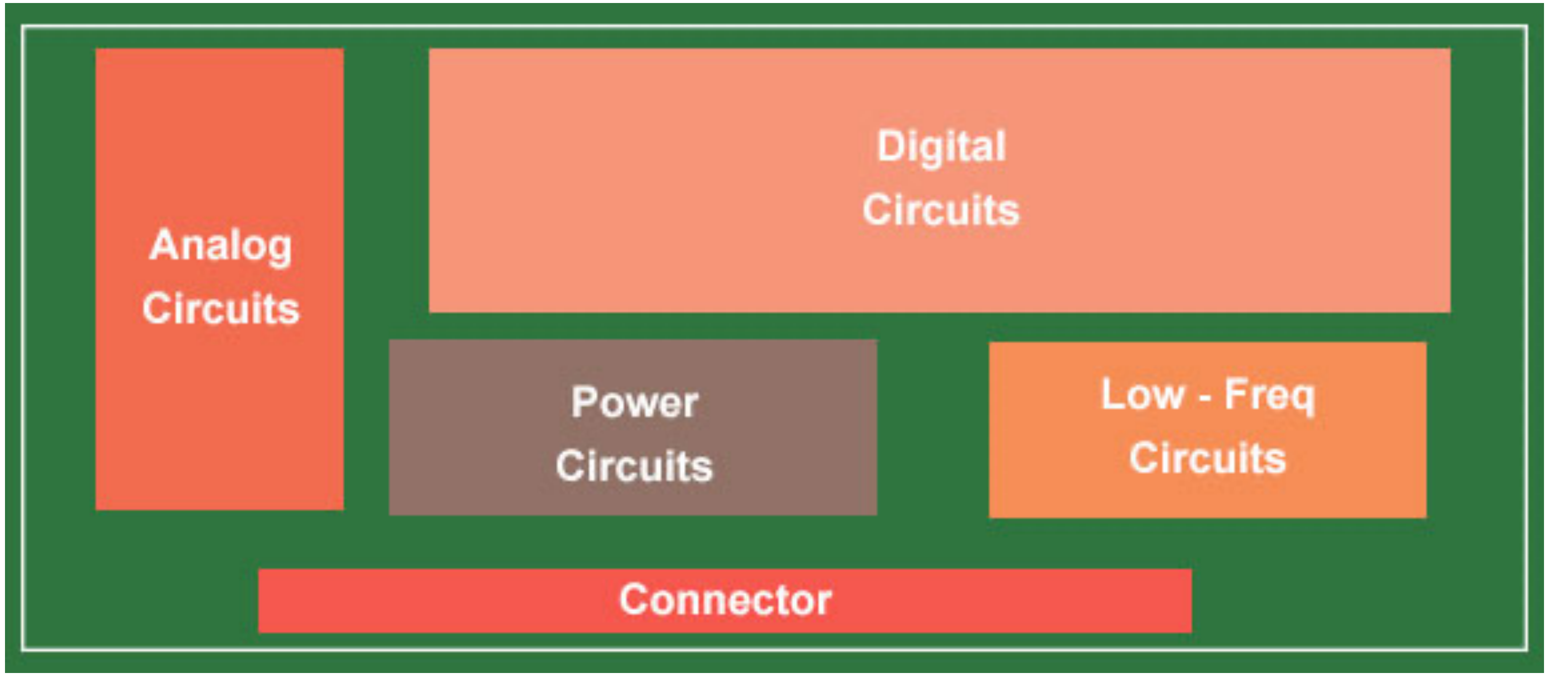
A microstrip hairpin filter followed by a low pass stub filter on a PCB in a 20GHz Agilent N9344C spectrum analyzer

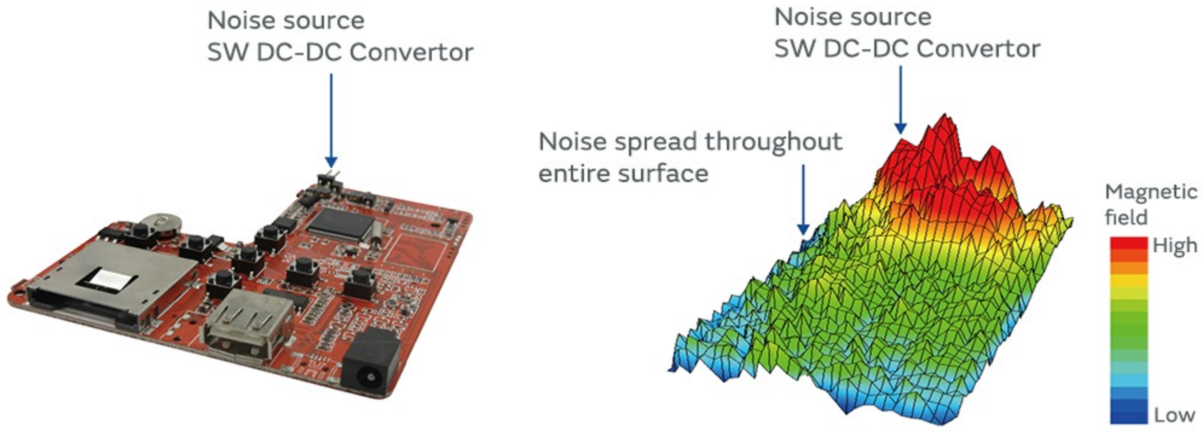


Stepped-impedance low-pass filter formed from alternate high and low impedance sections of line

PCB layouts and noise considerations

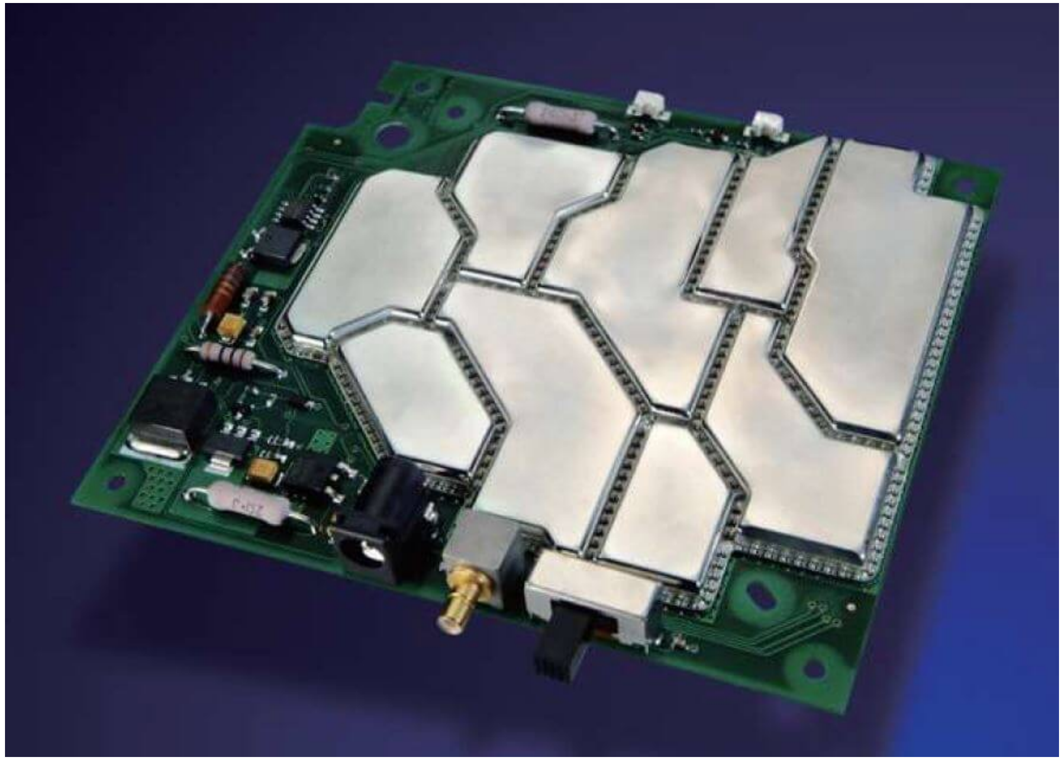


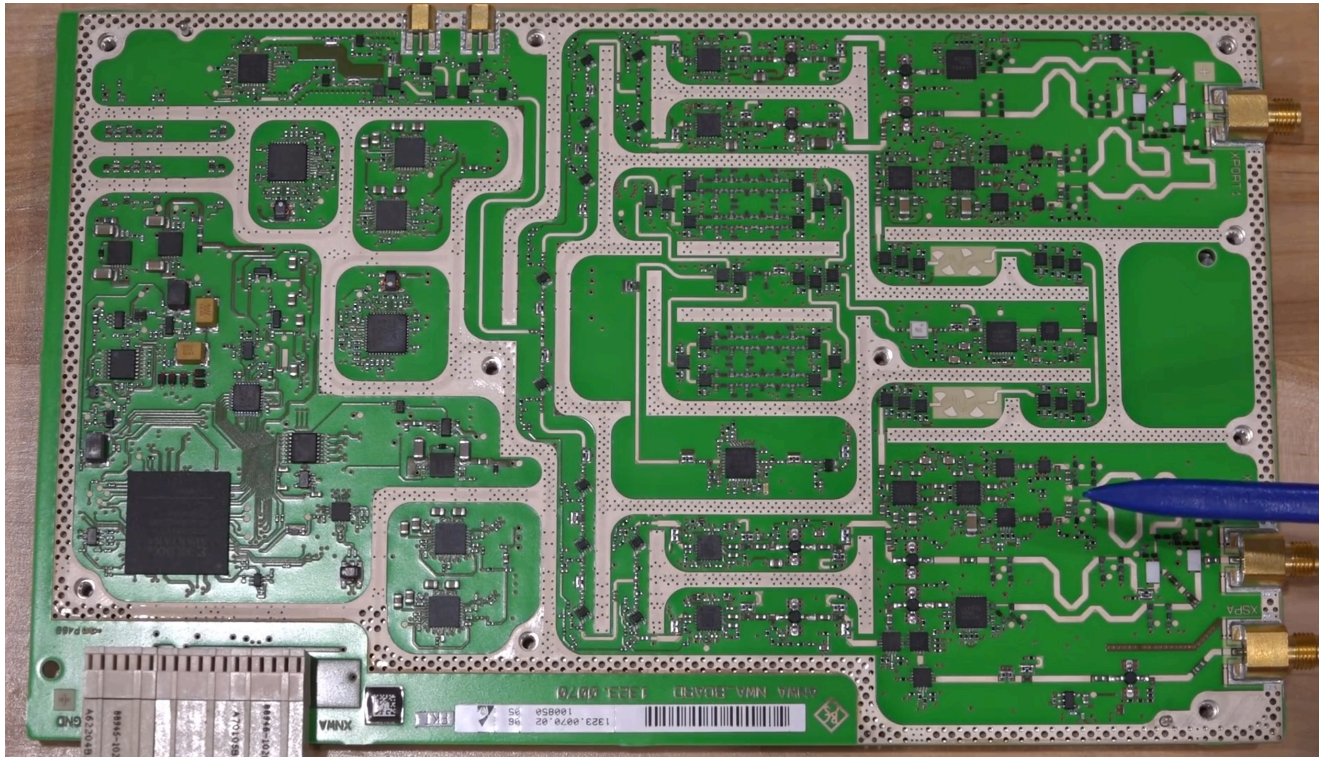




Caging

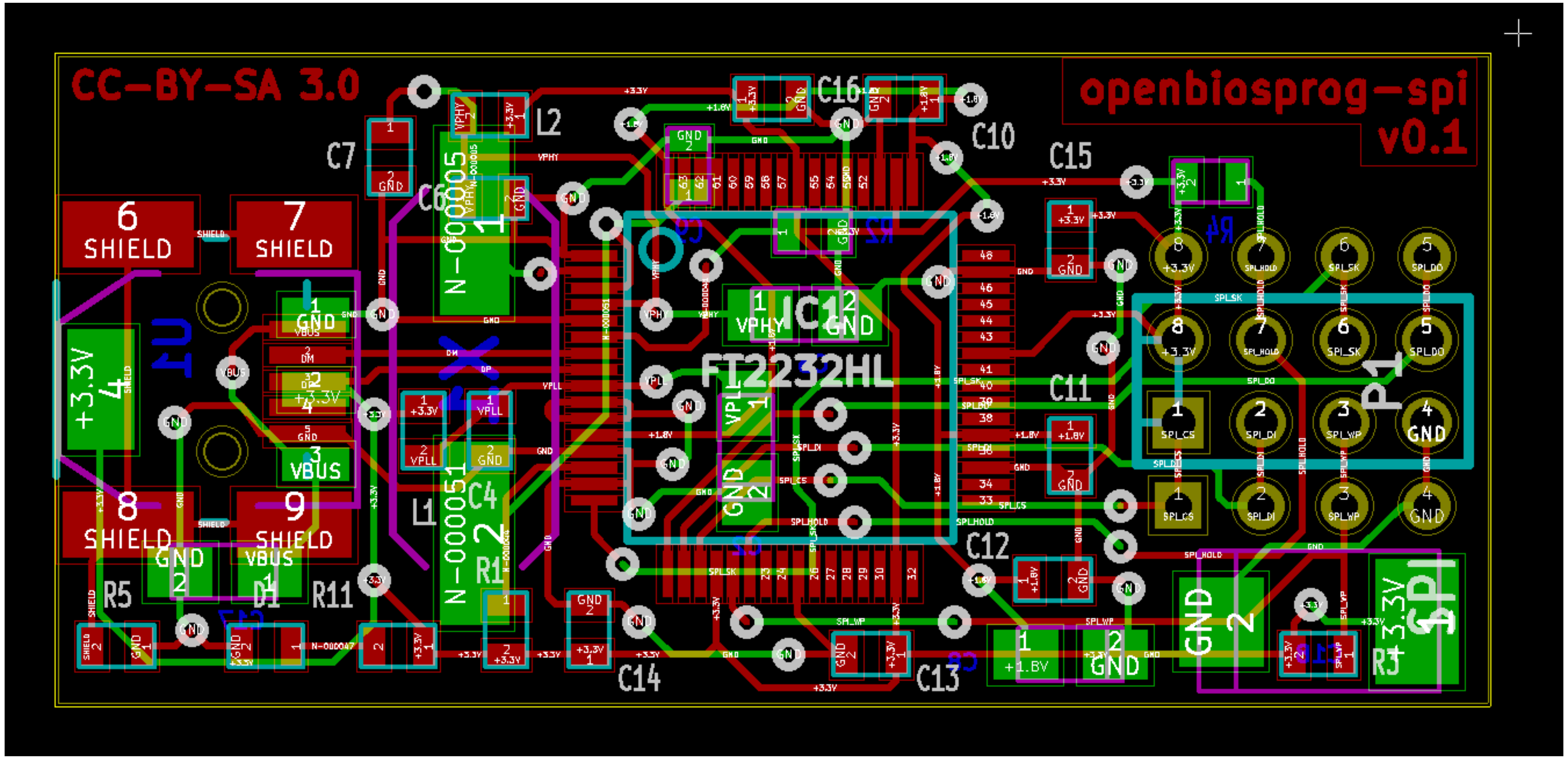
- Shielding cans or shielding cages are typically made of aluminum and are mounted on the PCB around specific groups of components.
- These small metal shields are then connected to ground, forming a Faraday cage around the components.
- These mechanical components are most often placed on a PCB as an EMI solution, either to help with passing susceptibility testing, radiated emissions testing, or both.





GERBER

“The Gerber format is an open vector format for printed circuit board (PCB) designs.”



Bill Of Materials

- **Reference Designator(s); i.e. “R1, R2, R3”**
- **Description; i.e. “10k 5% 0.125W Resistor”**
- *Manufacturer; i.e. “Vishay Dale”*
- *Manufacturers Part Number; it’s useful to link this to the datasheet. | i.e. “C0402C100J5GAC”.*
- *Distributor; once specified the distributor can be included. This will also help in tracking cost and leadtime as they may vary from one distributor to another. | i.e. “AVNet”*
- *Distributor Part Number; some PCBA manufacturing services require the customer to specify the distributor and orderable part number to save them time. It can take a significant amount of effort to shop around as to where to buy. | i.e. “C0402C100J5GACTU”*
- **Package; component size and shape | i.e. “SMD 0402”**
- **Qty; of each part per unit of product | i.e. “3”**

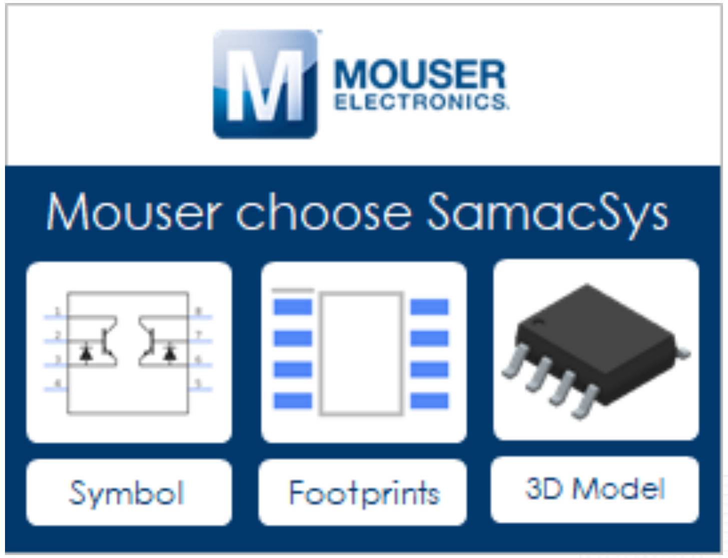
	A	B	C	D	E	H	I	J
3	Item #	Qty	Ref Des	Manufacturer	Manf part num	Description	Package	Type
4	1	1	U1	Atmel	ATMEGA328P-AUR	MCU	44-TQFP	SMT
5	2	1	U2	FTDI	FT231XS-R	USB to UART	20-SSOP	SMT
6	3	1	U2	Freescale	MMA8452QR1	Accelerometer	QFN16	SMT
7	4	1	Q1	Fairchild	KDT00030TR	Phototransistor	0603	SMT
8	5	1	Y1	Epson	NX5032GC-16MHZ-STD-CSK-6	16 MHz		SMT
9	6	3	D1, D2, D3	Osram	LY R976-PS-36	LED Yellow	0805	SMT
10	7	1	D4	Diodes, Inc	B0520LW-7-F	Schottky	0805	SMT
11	8	2	C1, C2	Yageo	CC0805DRNPO9BN8R0	8pf, 16V, 10%	0805	SMT
12	9	2	C3, C4	Yageo	CC0805KKX7R7BB105	1.0uf, 16V	0805	SMT
13	10	1	C5	Yageo	CC0805KRX7R9BB103	.01uf, 16V	0805	SMT
14	11	6	C6, C7, C8, C9, C13, C14	Samsung	CL21B104KOANNNC	.1uf, 16V	0805	SMT
15	12	2	C10, C11	Samsung	CL21C470JBANNNC	47pf, 16V	0805	SMT
16	13	1	C12	AVX	F931A476MAA	47uf, 16V		SMT
17	14	1	C26	Taiyo	LMK212BJ475KD-T	4.7uf, 10V	0805	SMT
18	15	2	L1, L2	Bourns	MH2029-300Y	30ohm ferrite	0805	SMT
19	16	1	R1	Stackpole	RMCF0805JT1M00	1M	0805	SMT
20	17	2	R2, R3	Stackpole	RMCF0805JT22R0	22 ohms	0805	SMT
21	18	3	R4, R5, R6	Stackpole	RMCF0805FT680R	680 ohm	0805	SMT
22	19	2	R7, R8	Rohm	MCR10ERTF2201	2.2K	0805	SMT
23	20	2	R9, R10	Stackpole	RMCF0805FT10K0	10K	0805	SMT
24	21	1	R11	Stackpole	RMCF0805JT4K70	4.7K	0805	SMT
25	22		S1, S2	C&K	PTS645SM43SMTR92 LFS	Momentary push		DNS
26	23	2	S3, S4	C&K	PTS645SM43SMTR92 LFS	Momentary push		SMT
27	24		J1	FCI	10118192-0001LF	USB microB		DNS
28	25	1	J2	FCI	10118192-0001LF	USB microB		SMT
29	26	1	B1	Linx	BC501SM	12mm Coin cell		SMT

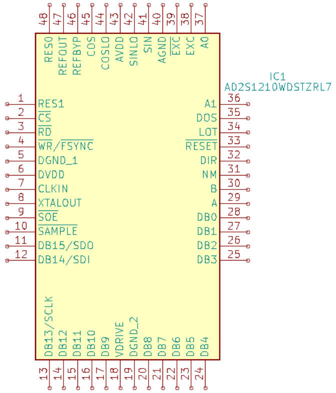
POS

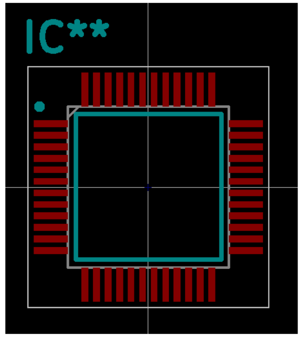
- Designator - Component Reference Designator (e.g. C1, L2, R3)
- Mid X/Mid Y - The X/Y coordinate of the component centroid. Generally in mm
- Layer - Top / Bottom, the board side where the component should be placed.
- Rotation - The rotation of the component given in degrees.
- Recommended File Format: .csv, .xls and .xlsx.

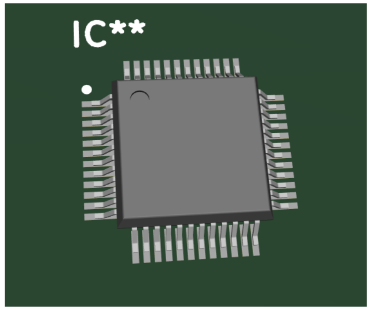
```
### Module positions - created on Monday, January 02, 2023 at 10:32:14 AM ###
### Printed by Pcbnew version (6.0.0-0)
## Unit = mm, Angle = deg.
## Side : bottom
# Ref     Val                    Package                          PosX       PosY       Rot   Side
BT1       Battery_Cell           CR2032-LCSC-C964760          -105.9500   -94.1500  135.0000  bottom
C1        0.1uF                  C_0805                        -93.3000   -94.8000  135.0000  bottom
D1        Schottky               D_SOD-323F                    -97.7000  -110.9000  -135.0000  bottom
D2        LED_RGBA               LED_Reverse_XZMDKCBDDG45S-9   -96.9000  -100.3000  125.0000  bottom
D4        LED_RGBA               LED_Reverse_XZMDKCBDDG45S-9   -92.9000  -102.8000  125.0000  bottom
D6        LED_RGBA               LED_Reverse_XZMDKCBDDG45S-9   -89.0000  -105.6000  125.0000  bottom
D8        LED_RGBA               LED_Reverse_XZMDKCBDDG45S-9   -85.0000  -108.5000  125.0000  bottom
J1        USB_C_Receptacle_USB2.0  USB_C_Power_Only_C2927027  -100.1000  -119.2000    0.0000  bottom
R1        5.1_kΩ                R_0805                       -101.4000  -108.4000  -135.0000  bottom
R2        5.1_kΩ                R_0805                       -103.4000  -111.1000  -90.0000  bottom
R3        100_Ω                 R_0805                        -83.1000  -101.7000  135.0000  bottom
R4        100_Ω                 R_0805                        -92.0000  -111.1000  -55.0000  bottom
R5        100_Ω                 R_0805                        -90.4000  -112.8000  -55.0000  bottom
SW1       SW_SPDT                Switch,_Side_-_LCSC_C963207  -110.1491  -110.6595   50.0000  bottom
SW2       SW_Push                500-0001-TS1187ACGB-(2_Pin)  -115.3000   -82.9000   45.0000  bottom
U1        ATtiny1614-SS          SOIC-14_3.9x8.7mm_P1.27mm     -88.6500   -90.1000  135.0000  bottom
## End
```

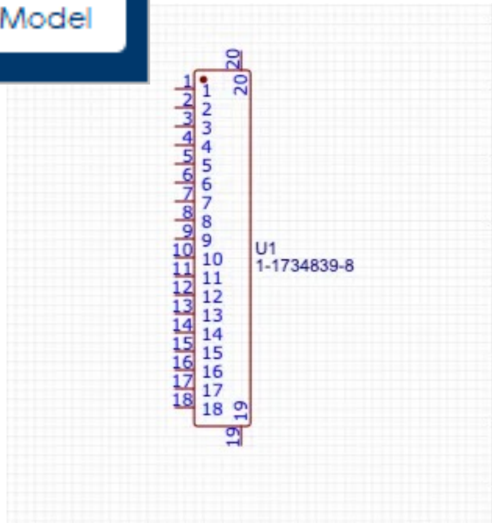
Symbol – Footprint and 3D model

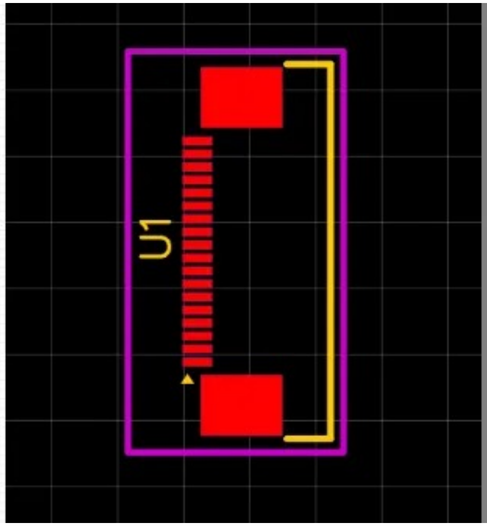


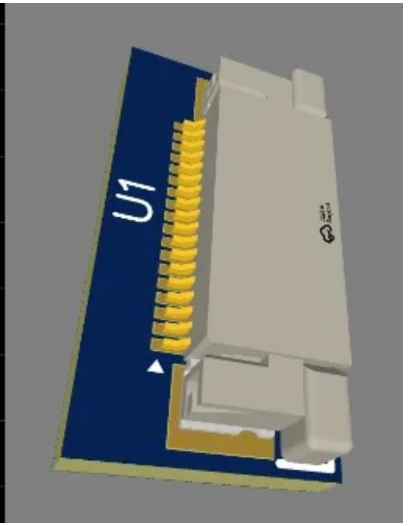




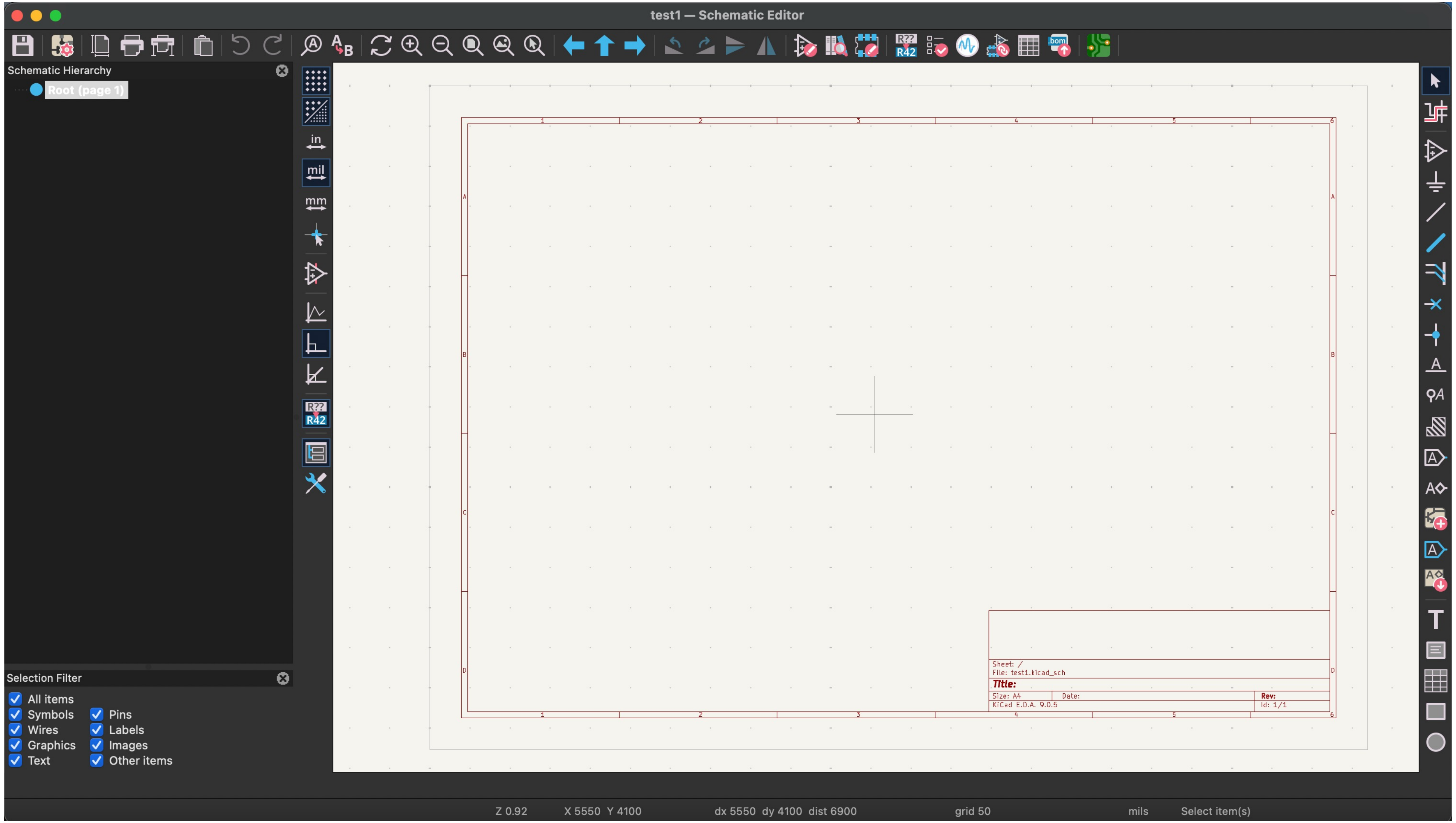








Schematic Design in KiCAD



PCB layout in KiCAD